



Human Computer Interfaces (HCI) Part 1

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Human Computer Interfaces (HCI)

<i>Smart Interfaces</i>		
Medical Engineering	Digital Signal\Image Processing	Machine Learning

Course outline

HCI (based on biological & Voice Signals)

- ❖ Introduction to HCI and Digital Signal processing Recap
- ❖ EOG based interfaces
- ❖ Brain Computer interfaces (BCIs)
- ❖ ECG based interfaces (authentication)
- ❖ EMG based interfaces
- ❖ Voice based interfaces

Assessments

- ❖ Final Exam: 50
- ❖ Midterm: 15
- ❖ Practical tasks: 10
- ❖ Project: 25

References (books)

- ❖ Brain-Computer Interfacing

An introduction

Rajesh P. N. Rao

- ❖ Digital Signal Processing

Principles, Algorithms, and Applications

Third Edition

John G. Proakis, Dimitris G. Manolakis



- Digital Signal Processing

A practical approach

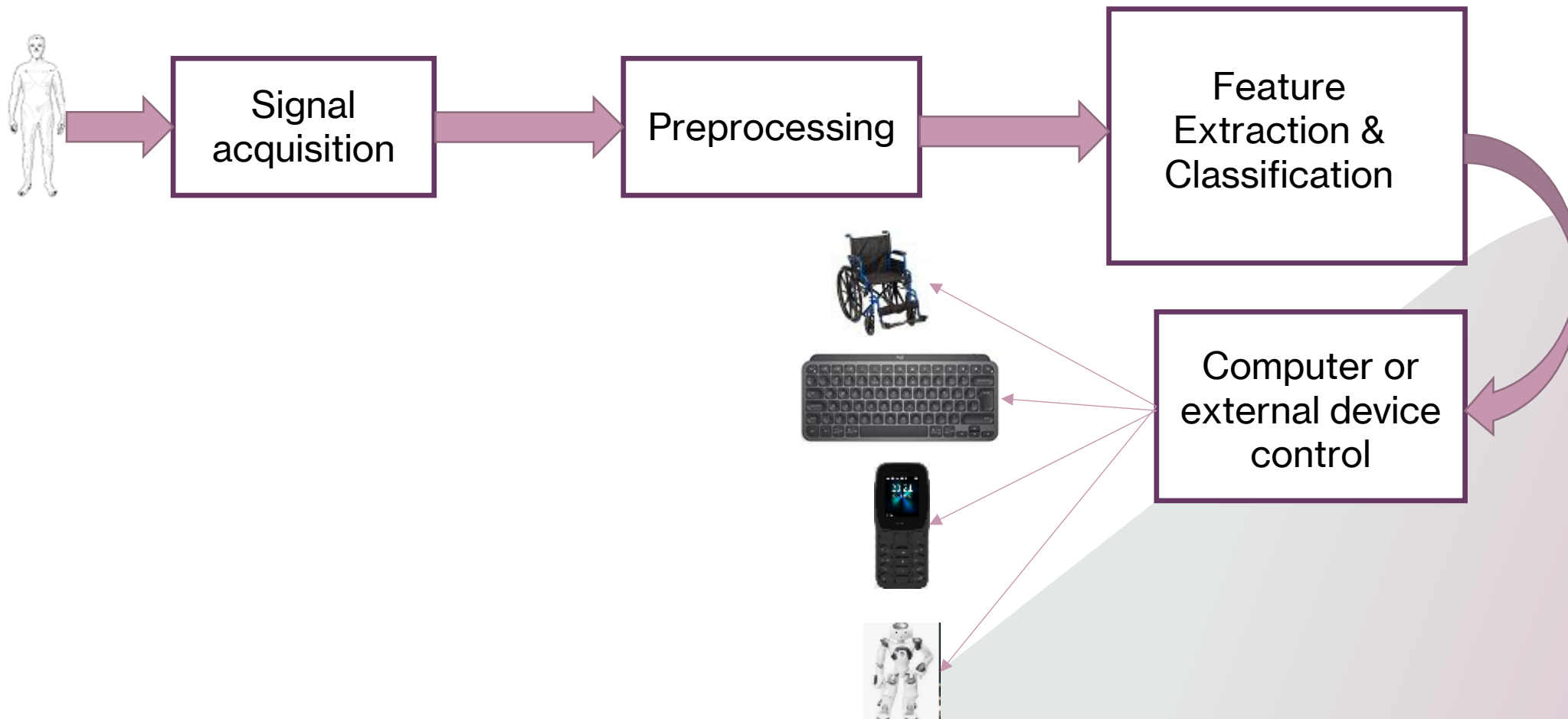
Third Edition

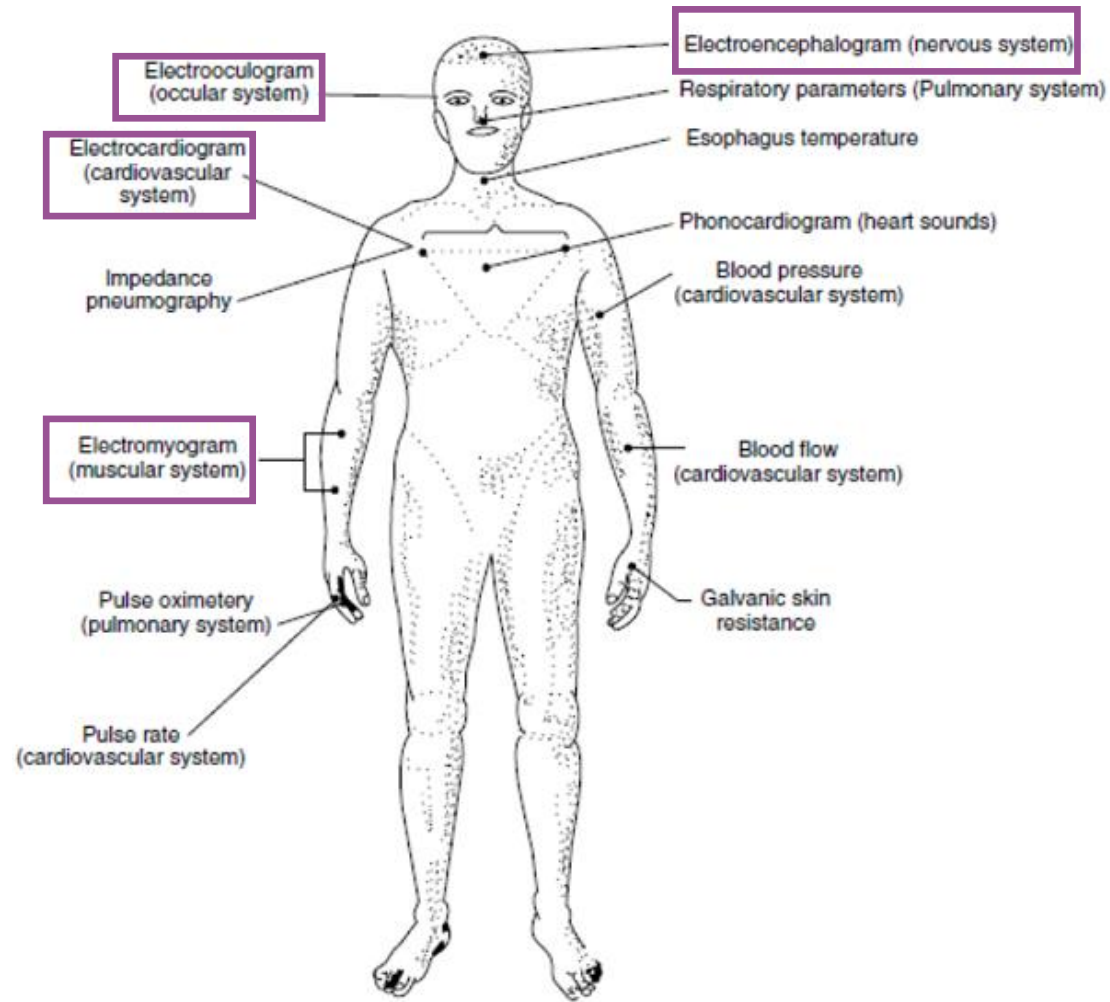
Emmanuel C. Ifeathor, Barrie W. Jervis

Part outline

- ❖ Introduction to HCI
- ❖ Digital Signal Processing-Recap

Human Computer Interface (HCI)





Biological Signals

Data Acquisition

➤ **Biological Signals**

- Electroencephalogram (EEG)
- Electrooculography (EOG)
- Electrocardiogram (ECG)
- Electromyography (EMG)

Biological signals are captured using specific devices, filtered, digitized and then transferred to computer directly (ex: via Bluetooth or USB) or through Arduino.

➤ **Human Voice**

Voice is usually recorded through mic as audio files

➤ **Images**

Images are captured by cameras

➤ **Preprocessing**

Signals are enhanced using suitable digital signal processing techniques.

➤ **Feature Extraction and Classification**

Extract suitable features for task in hand and then use machine learning approach (ex: ANN, SVM,...etc) or use a deep learning model (ex: CNN, RNN,...) for both feature extraction and classification.

➤ **Computer or Device Control**

The needed action is executed by the computer or in case of external device the action can be passed to this device through Arduino

Motivation for HCs

- Help disabled people to interact with their environment as much as possible without needing help.
- Rehabilitation

Digital Signal Processing...Recap

- ❖ Signals
- ❖ Sampling and quantization
- ❖ Digital Fourier Transform (DFT)
- ❖ Discrete cosine transform (DCT)
- ❖ Convolution, Cross-Correlation and Auto-correlation
- ❖ Filtering
- ❖ Time versus Frequency Domains
- ❖ Short Term Fourier Transform (STFT)
- ❖ Wavelet Transform
- ❖ Enhancing Signals in Time and Frequency Domains

Signals



Signals



Signal is defined as any physical quantity that varies with time, space, or any other independent variable or variables. Mathematically, we describe a signal as a function of one or more independent variables.

Ex: speech , biological signals, image, video, and radar signals

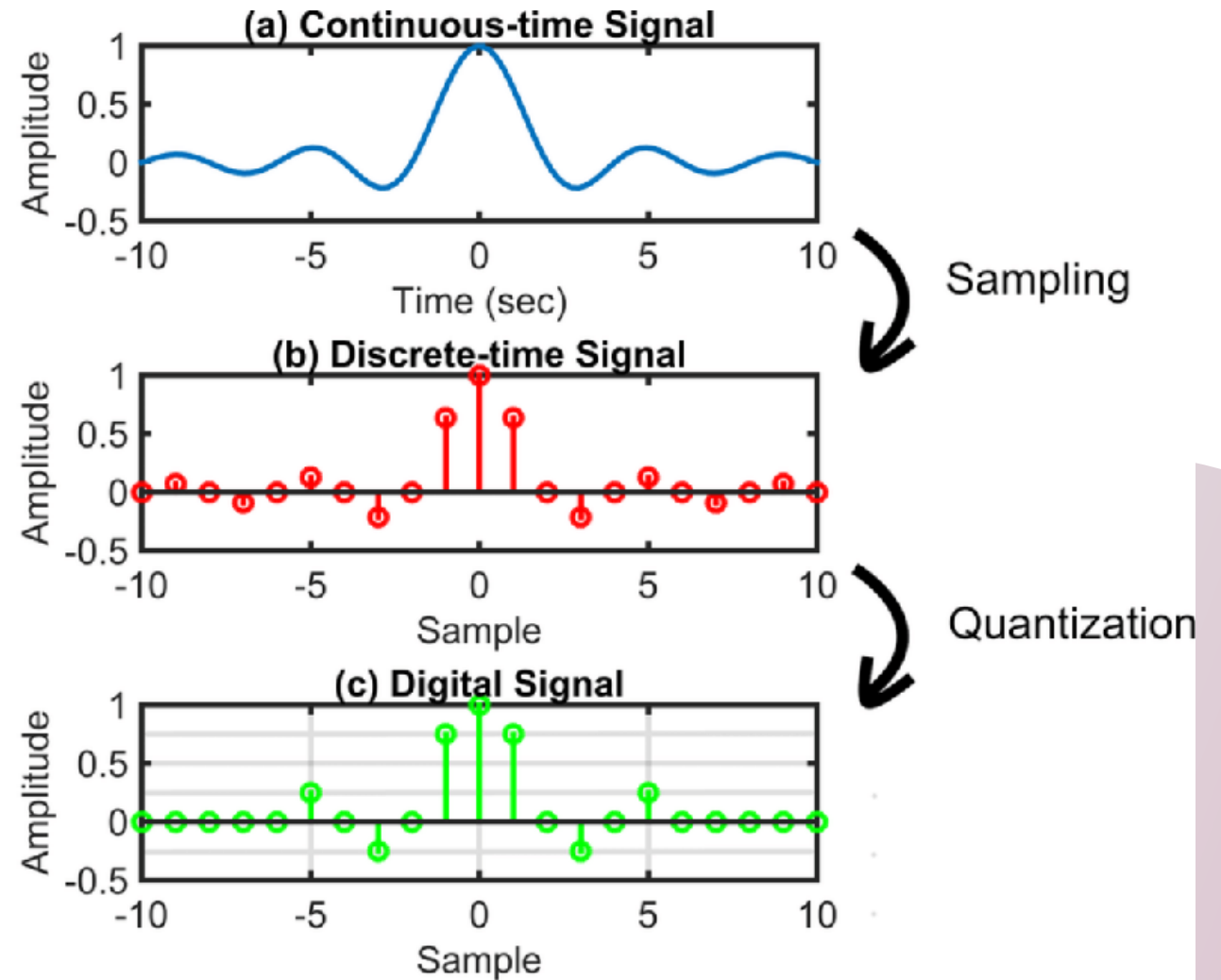
Few signals can be represented mathematically

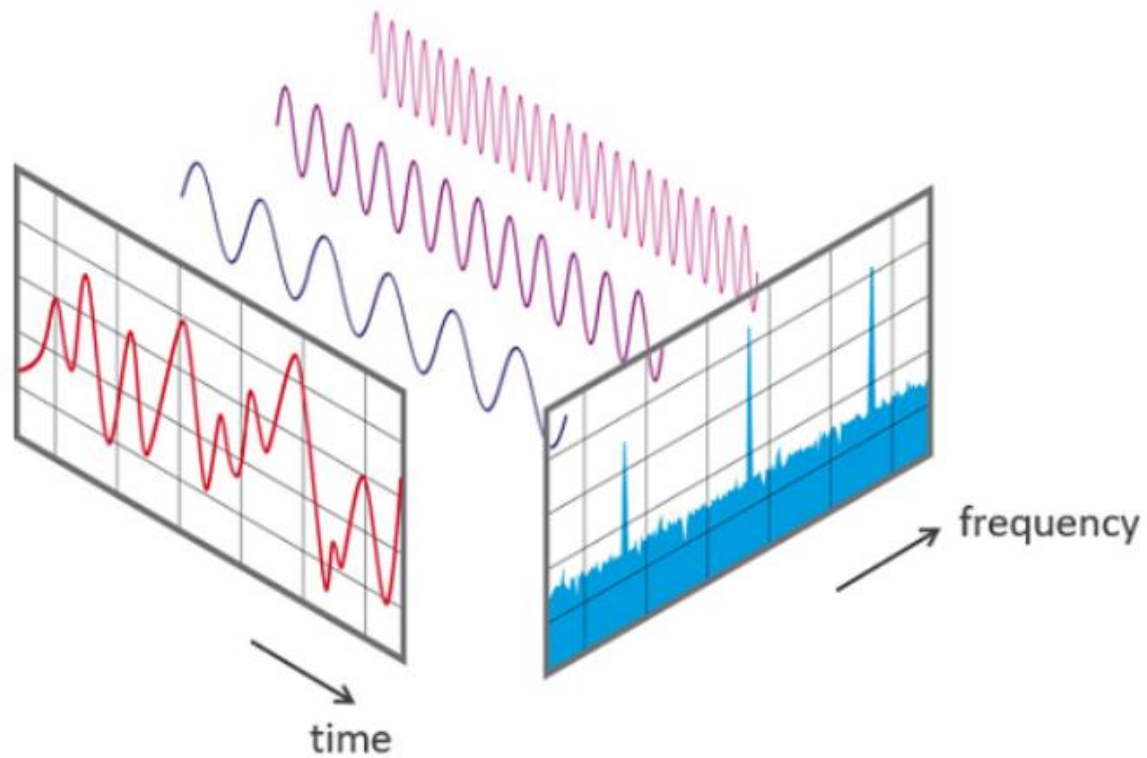
Signals can be:

- Analog, discrete or digital
- One or multidimensional (ex: image)
- One or multichannel (ex: EEG, ECG, etc..)
- Deterministic or random



Sampling & Quantization





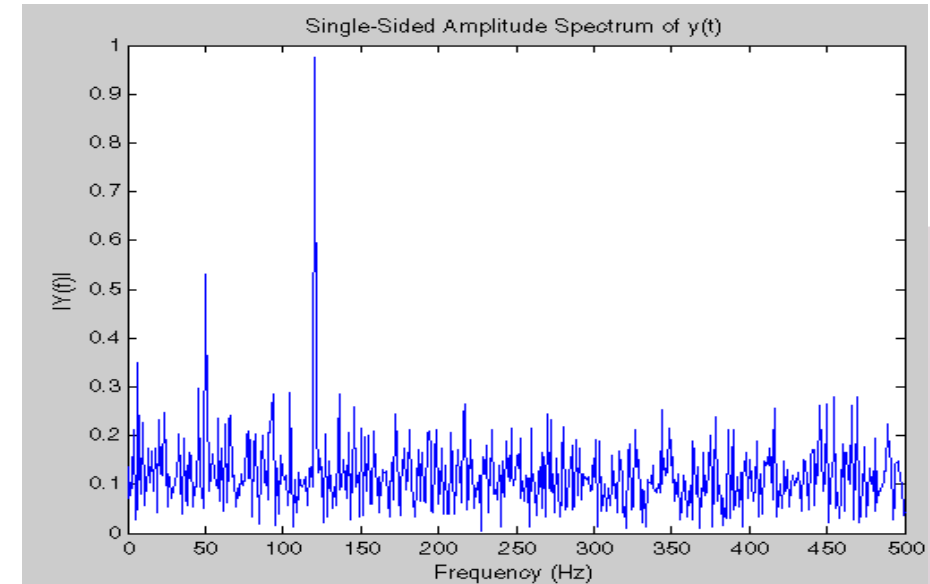
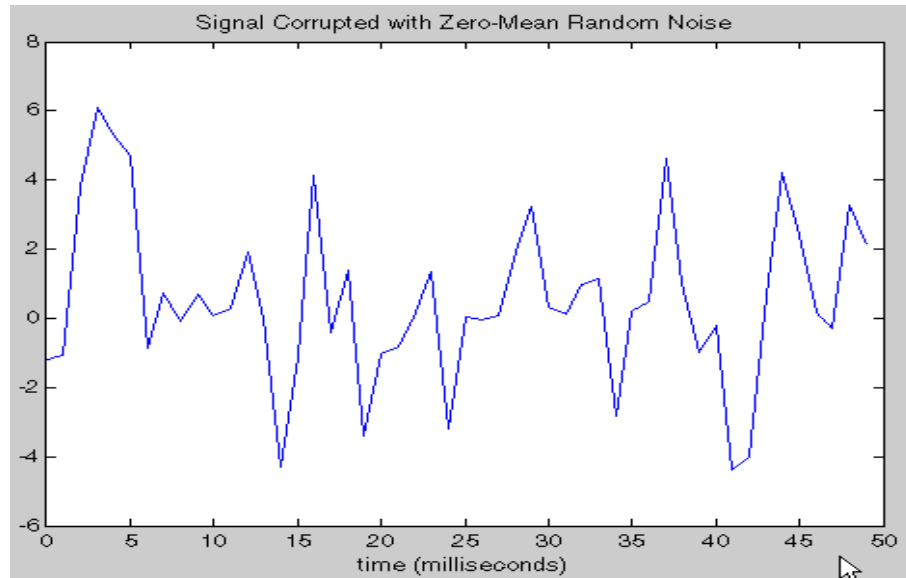
Digital Fourier Transform (DFT)

Fourier Transform is to decompose a signal into a weighted sum of sine and cosine waves of different frequencies.

Discrete Fourier Transform (DFT)

Assume that a waveform has been sampled at regular time intervals T to produce the sample sequence $\{x(nT)\} = x(0), x(T), \dots, x[(N-1)T]$ of N sample values, where n is the sample number from $n=0$ to $n=N-1$. The DFT of $x(nT)$ is then defined as the sequence of complex values $\{X(k\Omega)\} = X(0), X(\Omega), \dots, X[(N-1)\Omega]$ in the frequency Domain where Ω is the first harmonic frequency given by $\Omega = 2\pi/NT$. Thus, the $X(k\Omega)$ have real and imaginary components in general so that for the k th harmonic.

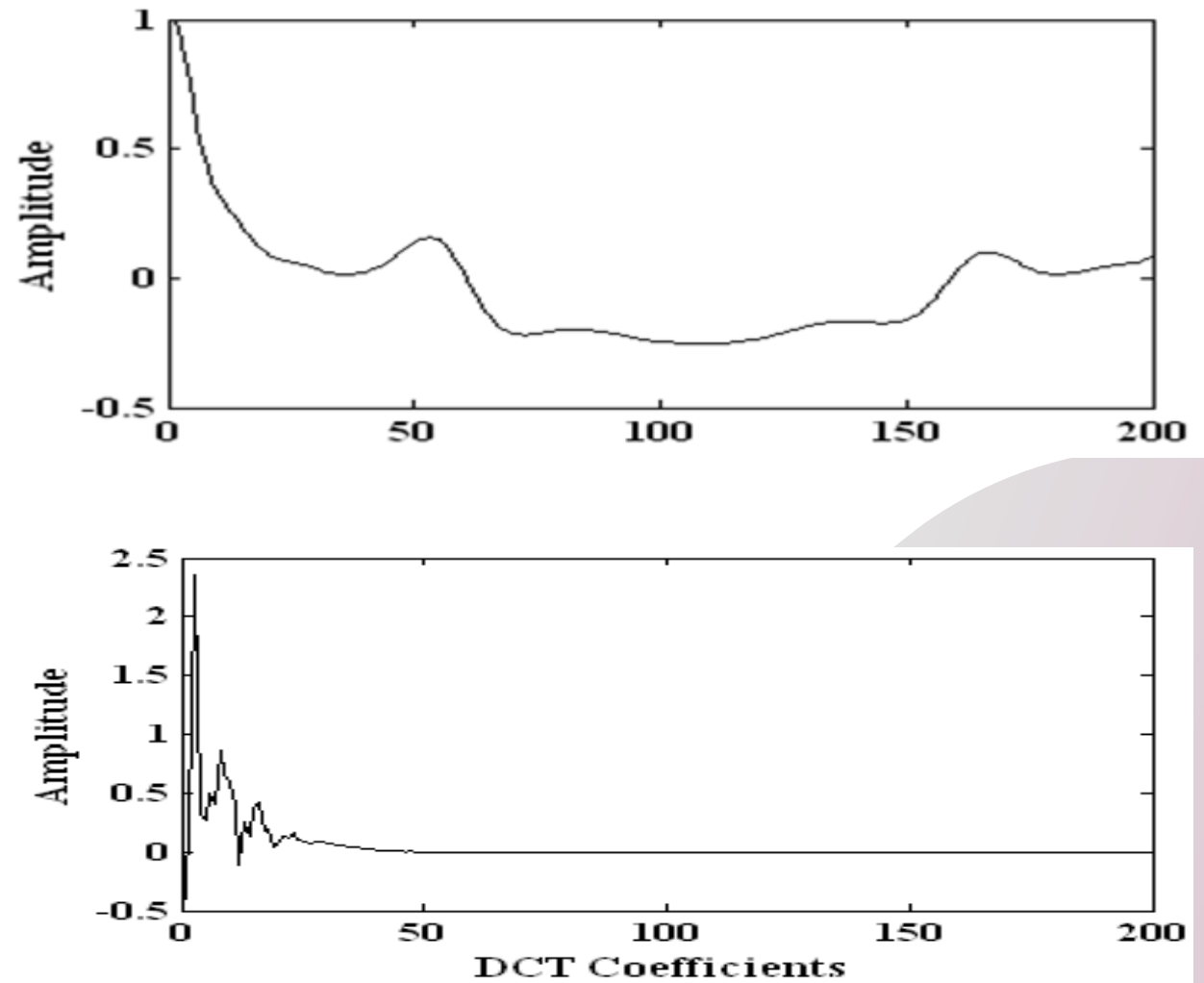
Discrete Fourier Transform (DFT)



Fast Fourier Transform (FFT)

FFT is a more efficient way of computing the DFT. It runs approximately in $N \log(N)$ which can result in huge saving in computation for large N compared to DFT which needs N^2 . The most common FFT algorithm, the Cooley-Tukey algorithm, uses a “divide and conquer” strategy and recursively breaks down a DFT into many smaller DFTs. Most signal processing packages come with an FFT implementation, making FFT the most used method for transforming a time-varying signal to the frequency domain.

Discrete Cosine Transform (DCT)



Discrete Cosine Transform (DCT) is frequently used in lossy data compression applications, such as the JPEG image format.

❖ The property of the DCT that makes it quite suitable for compression is its high degree of "spectral compaction;" at a qualitative level, a signal's DCT representation tends to have more of its energy concentrated in a small number of coefficients when compared to other transforms like the DFT.

❖ This is desirable for a compression algorithm; if you can approximately represent the original (time- or spatial-domain) signal using a relatively small set of DCT coefficients, then you can reduce your data storage requirement by only storing the DCT outputs that contain significant amounts of energy (ex: JPG images).

Convolution, Cross-Correlation and Auto-Correlation

Basic operations for signal processing

Convolution

This formula that gives the response $y(n)$ of the LTI system as a function of the input signal $x(n)$ and the unit sample (impulse) response $h(n)$ is called a convolution sum. We say that the input $x(n)$ is convolved with the impulse.

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

Cross-Correlation

It is a process of measuring the degree of similarity (dependency) between two data sets (signals). Given two data sequences x_1 and x_2 of length N , the correlation between them is computed by taking the summation of the product of the corresponding pairs of points.

$$r_{12}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)x_2(n+j) \quad \text{Lag } j=0, 1, 2, \dots, N$$

Normalizing cross-correlation $[-1, 1]$

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N} \left[\sum_{n=0}^{N-1} x_1^2(n) \sum_{n=0}^{N-1} x_2^2(n) \right]^{1/2}}$$

Auto-Correlation

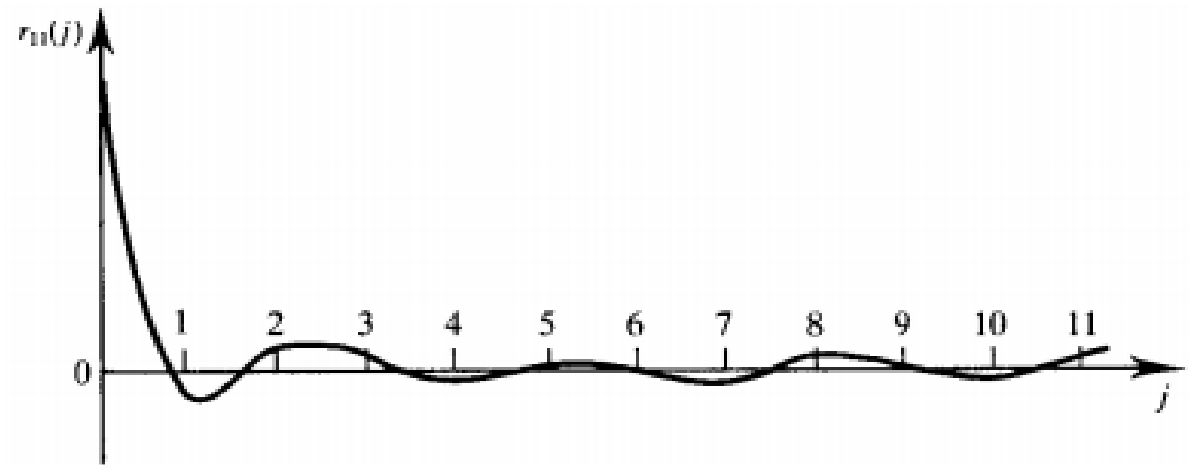
❖ The auto-correlation function measures the correlation of a signal $x(n)$ with itself shifted by some time delay j :

$$r_{11}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)x_1(n + j)$$

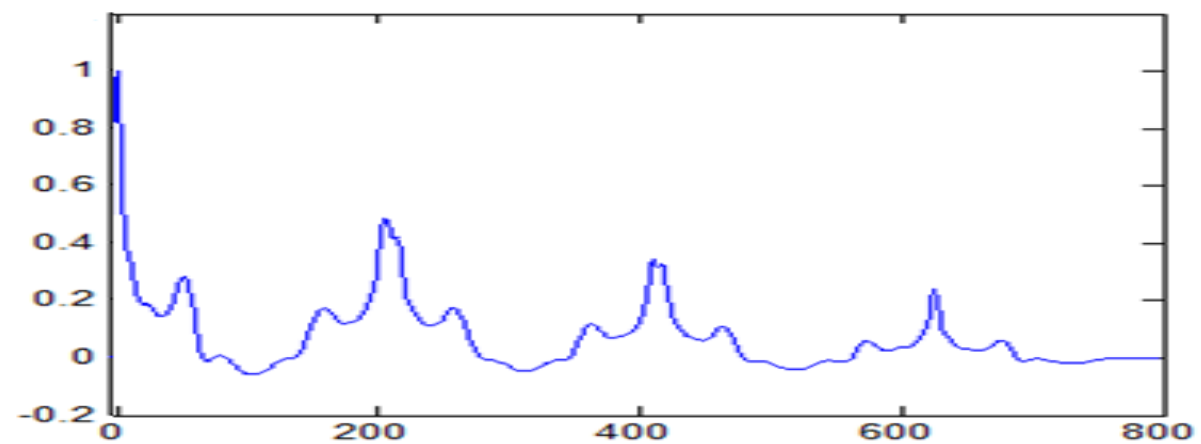
❖ Its maximum value is achieved at lag $=0, \dots, N$ (complete overlap with itself) where N is the sequence length.

❖ The auto-correlation function can be used to detect repeats or periodicity in a signal or if it is pure random. Moreover, it can identify signals that are embedded in noise.

Auto-Correlation

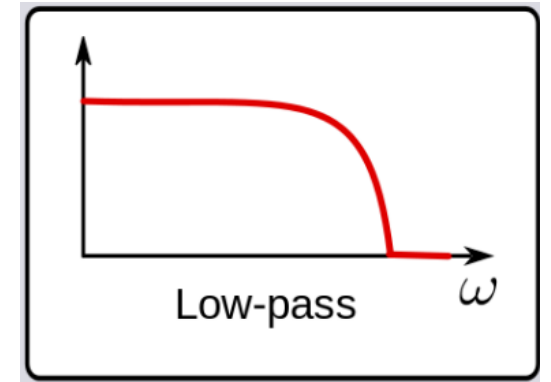
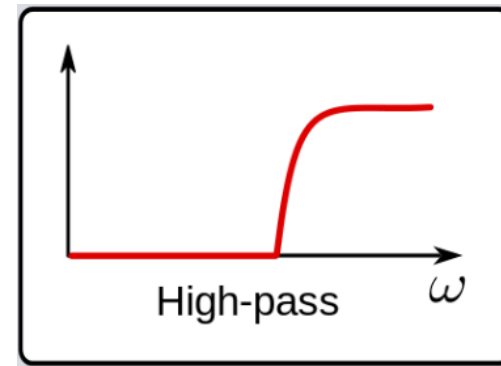
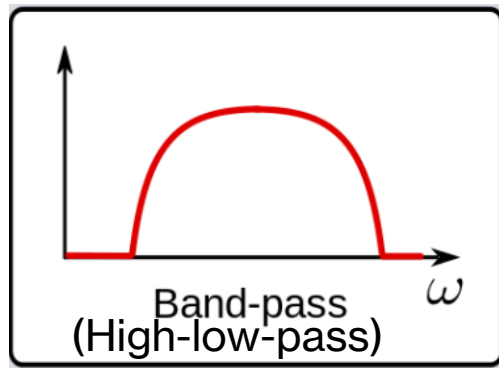
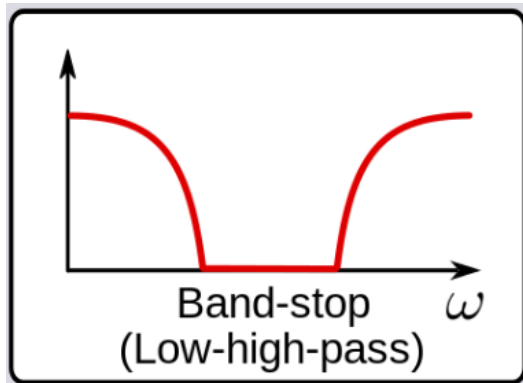


Autocorrelation function of a random waveform.



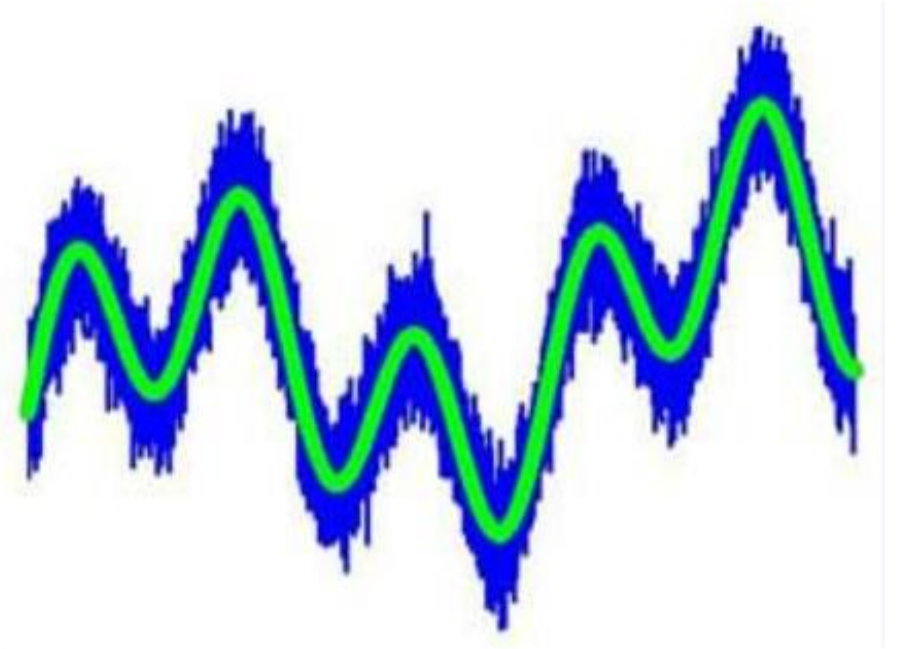
Periodic signal

Filtering

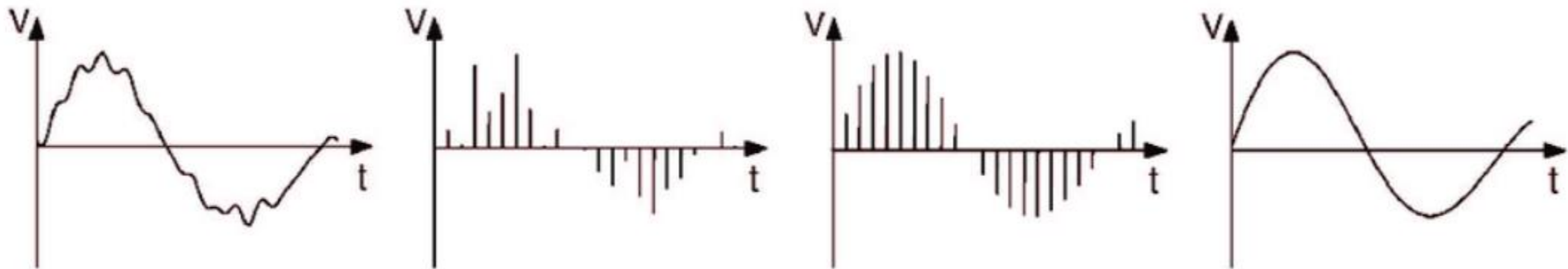


Filtering

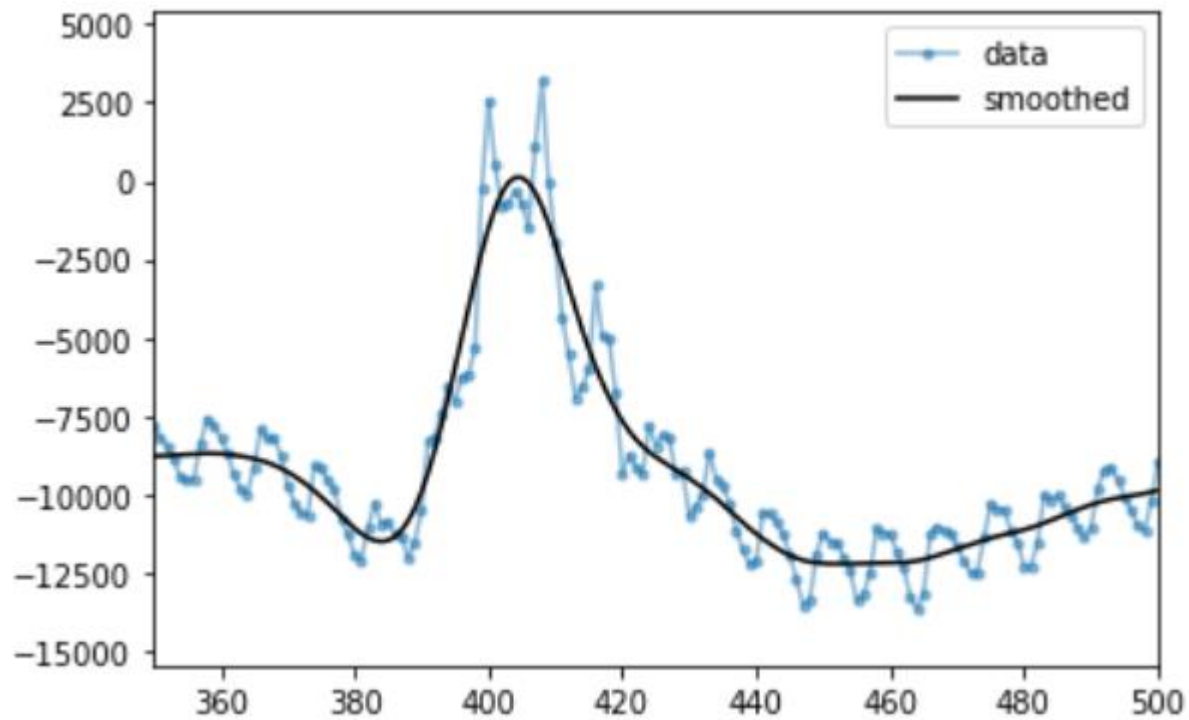
- ❖ A filter is essentially a system or network that selectively changes the wave-shape, amplitude-frequency and /or phase-frequency characteristics of a signal in a desired manner.
- ❖ Common filtering objectives are to improve the quality of a signal (for example, to remove or reduce noise) or to extract information from signals.
- ❖ A digital filter is a mathematical algorithm implemented in hardware and/ or software that operates on a digital input signal to produce a digital output signal for the purpose of achieving a filtering objective.



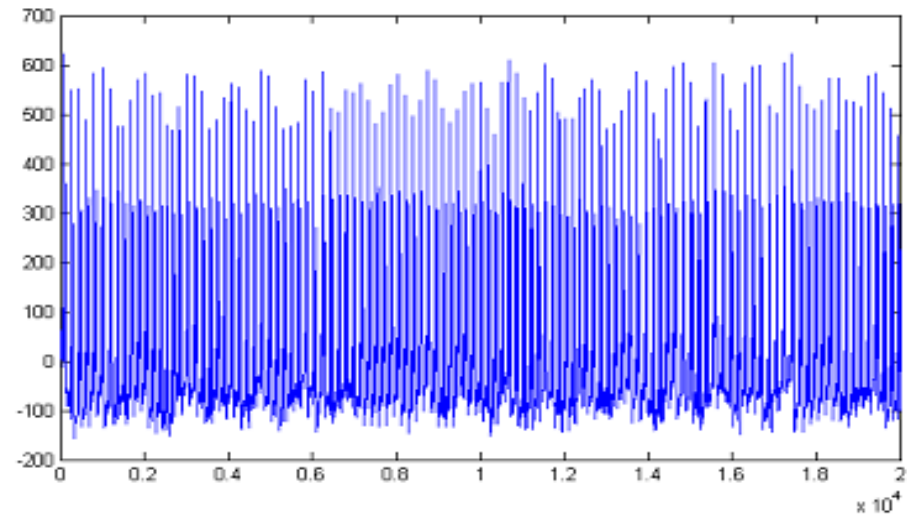
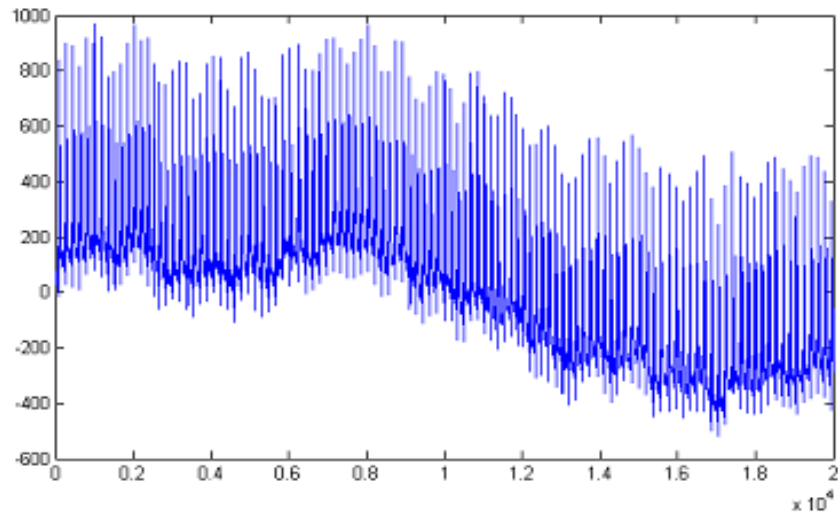
Digital Low Pass Filter (smoothing) example



Filtering (smoothing example)



Filtering (baseline wandering)



FIR versus IIR Filters

Digital filters are broadly divided into two classes finite impulse response (FIR) and infinite impulse response (IIR) filters. Both types can be represented by its impulse response sequence $h(k)$. The input and output signals to the filter are related by convolution some.

$$y(n) = \sum_{k=0}^{\infty} h(k)x(n - k) \quad \text{IIR}$$

$$y(n) = \sum_{k=0}^{N-1} h(k)x(n - k) \quad \text{FIR}$$

FIR versus IIR Filters

- ❖ For IIR filters, the impulse response is of infinite duration whereas for FIR it is of finite duration. In practice, it is not feasible to compute the output of the IIR filter because the length of its impulse response is too long (infinity). Thus, IIR filtering equation is expressed in a recursive form:

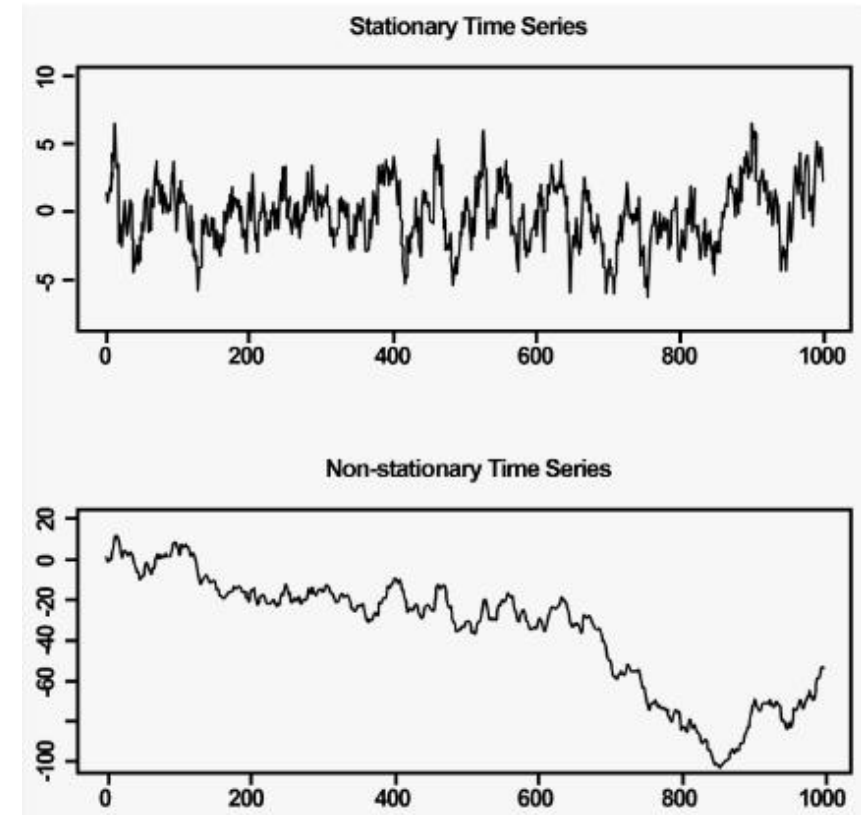
$$y(n) = \sum_{k=0}^{\infty} h(k)x(n-k) = \sum_{k=0}^N a_k x(n-k) - \sum_{k=1}^M b_k y(n-k)$$

- ❖ The a_k and b_k are the coefficients of the filter. Computing the values of $h(k)$ for FIR or a_k and b_k for IIR are often very important objectives of most filter design problems.

Time versus Frequency Domains

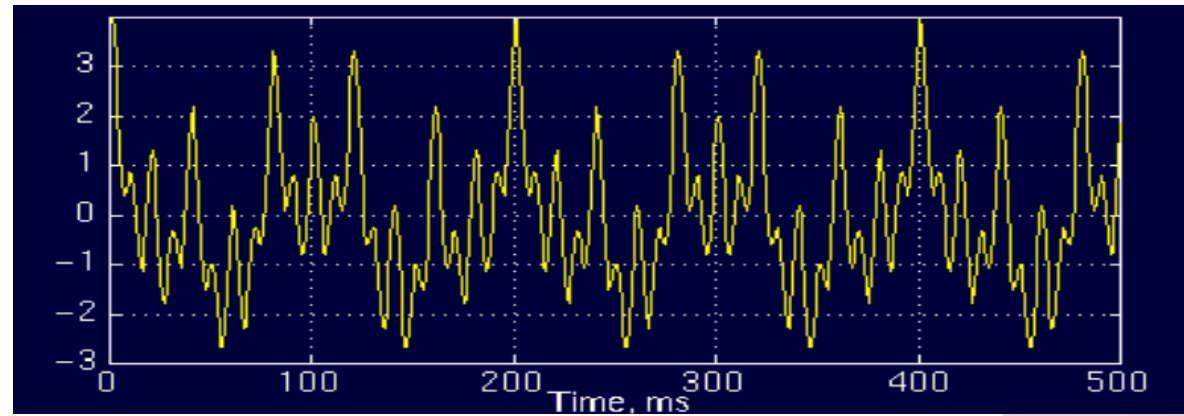
- Time domain: All time information (samples) is available but no information about frequency components.
- Frequency domain : Frequency components are known but no information about time.
- Do we need sometimes information about both time & frequency ?
YES, when we need to know when a specific frequency occurs since it doesn't exist all the time.

Stationary versus non-Stationary Signals

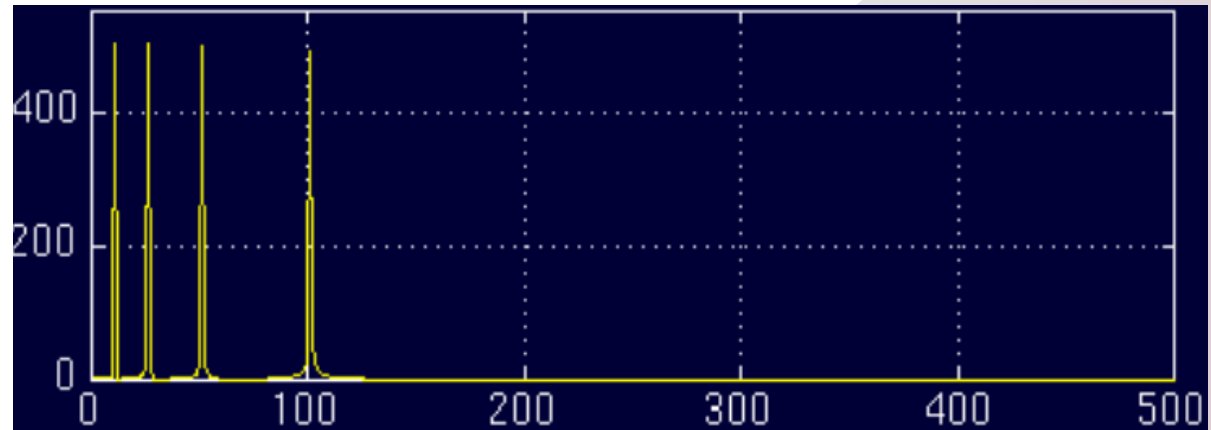


Stationary Signals

Signal

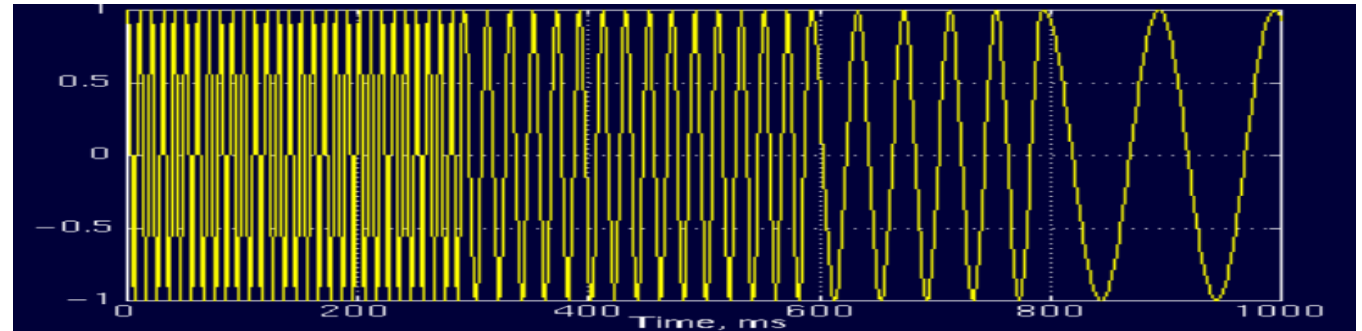


DFT

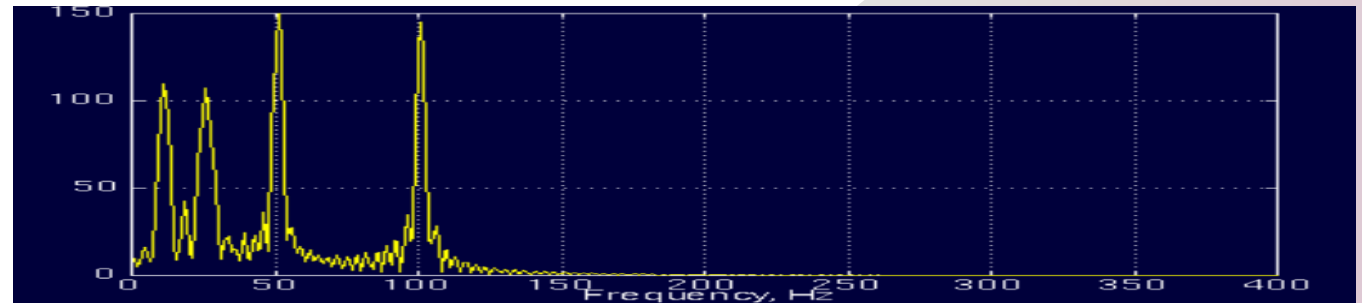


Non-Stationary Signals

Signal



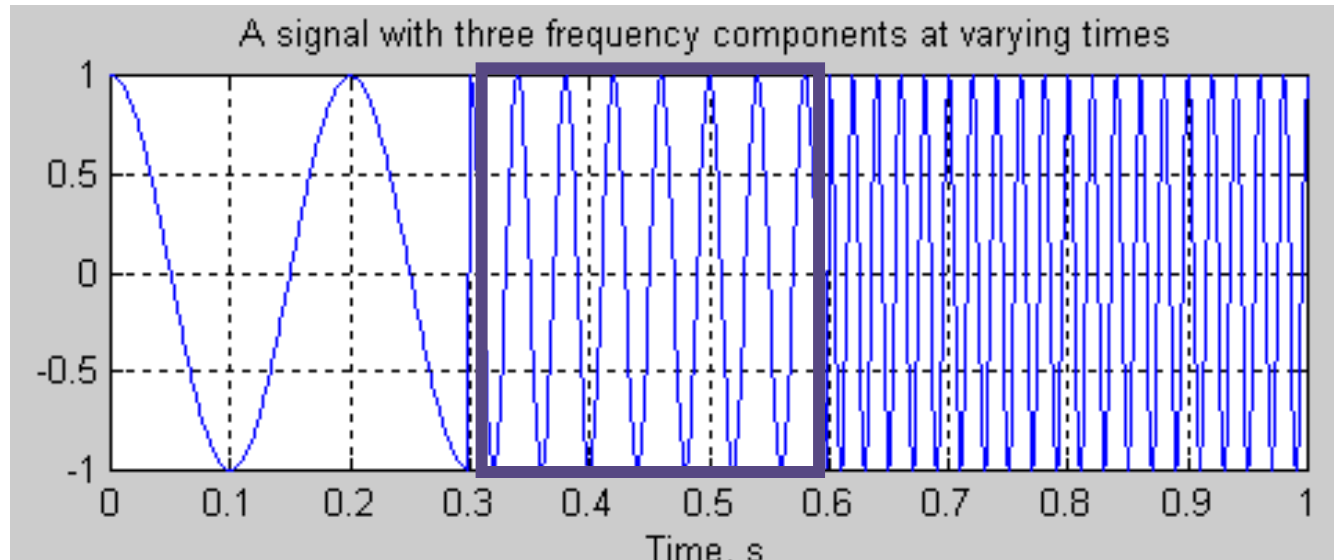
DFT



The spectrum is almost the same as the spectrum of the previous signal !!! Do the frequency components appear all the time? No
At what times (or time intervals), do these frequency components occur?
If you need the answer for such question, then DFT is not the right transform to use ?

Short Term Fourier Transform (STFT)

Can we assume that some portion of a non-stationary signal is stationary? **Yes**

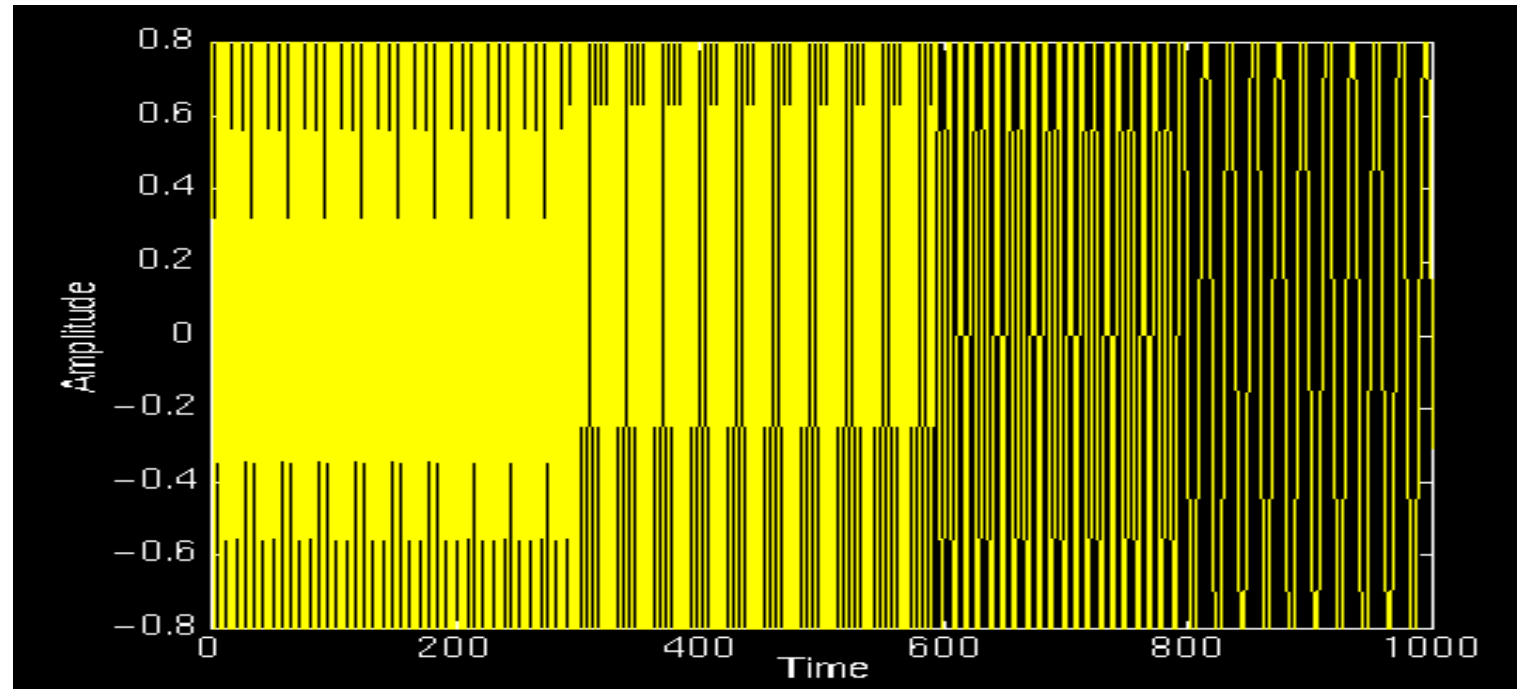


STFT algorithm

Steps :

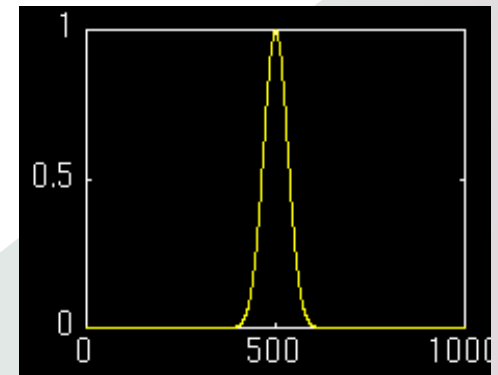
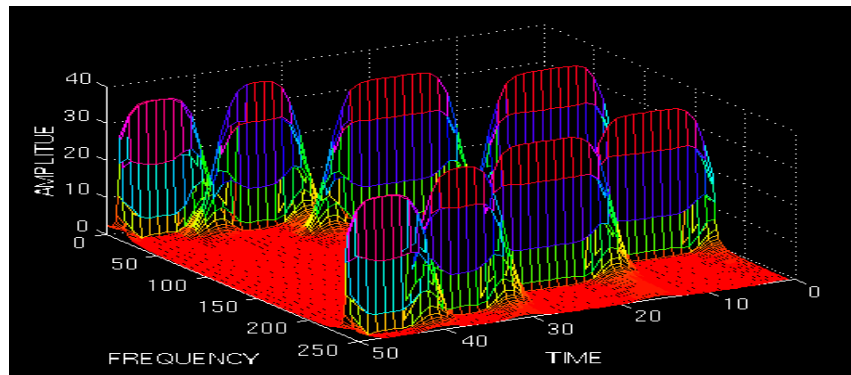
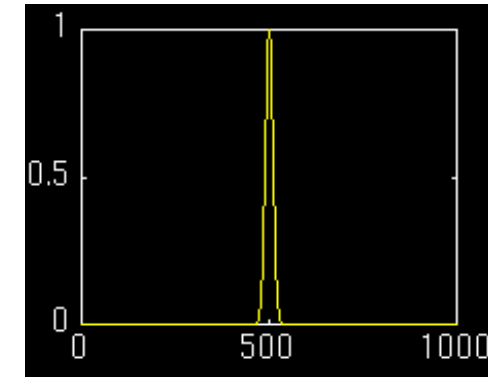
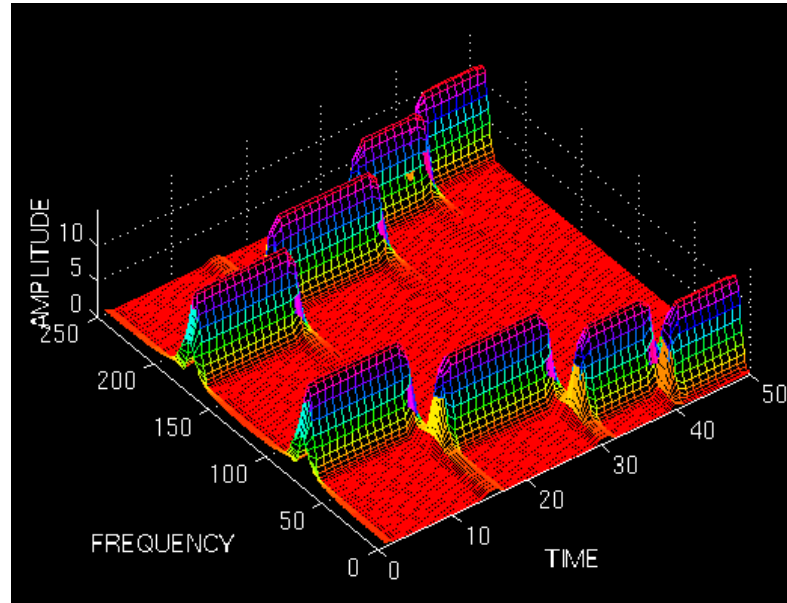
- (1) Choose a window function of finite length**
- (2) Place the window on top of the signal at $t=0$**
- (3) Truncate the signal using this window**
- (4) Compute the FT of the truncated signal, save results.**
- (5) Incrementally slide the window to the right**
- (6) Go to step 3, until window reaches the end of the signal.**

STFT algorithm (example)

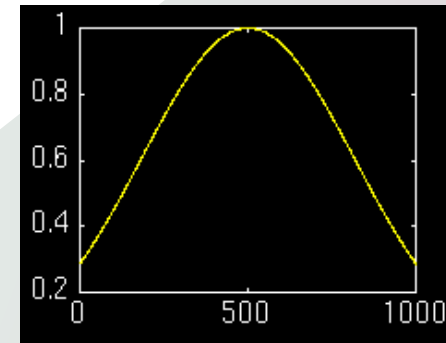
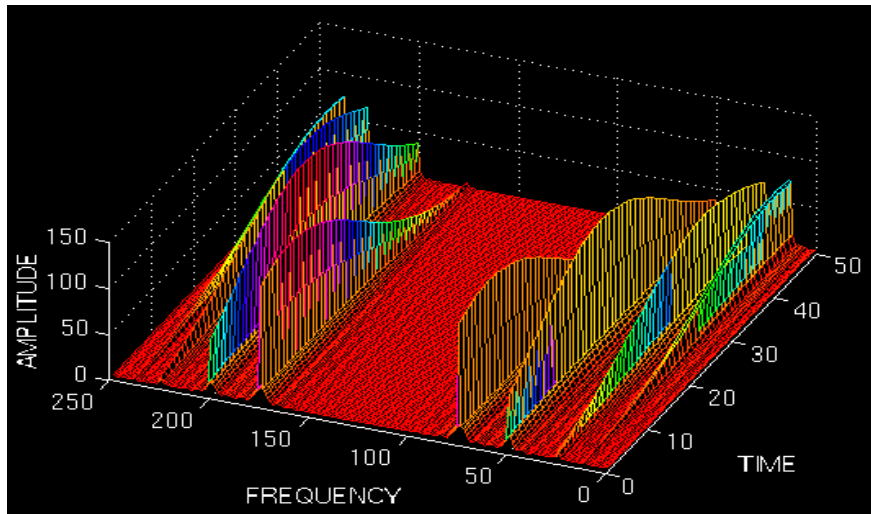
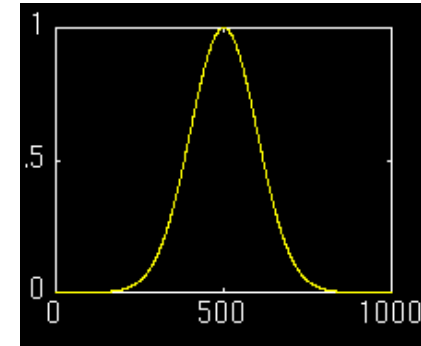
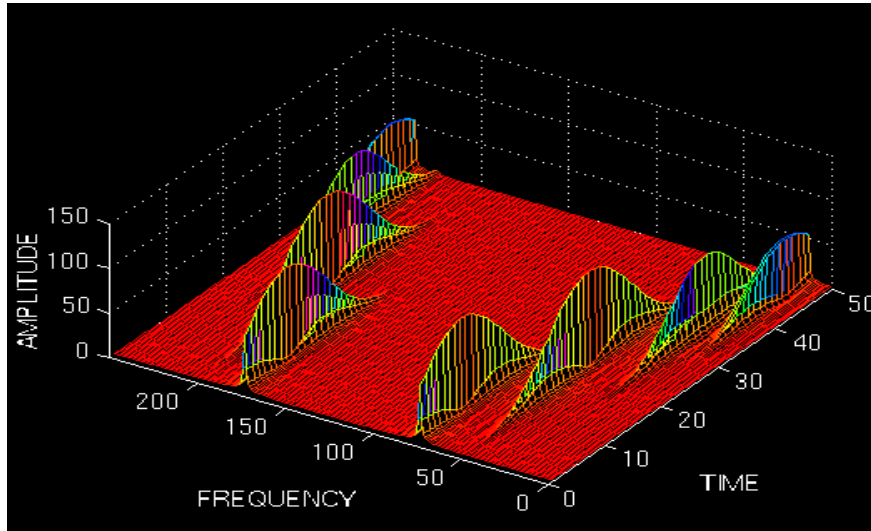


[0 – 300] ms → 75 Hz sinusoid
[300 – 600] ms → 50 Hz sinusoid
[600 – 800] ms → 25 Hz sinusoid
[800 – 1000] ms → 10 Hz sinusoid

STFT algorithm (example)



STFT algorithm (example)



Heisenberg (or Uncertainty) Principle

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

Time resolution: How well two spikes in time can be separated from each other in the transform domain.

Frequency resolution: How well two spectral components can be separated from each other in the transform domain

Δt and Δf cannot be made arbitrarily small!

Wavelet Transform (WT)

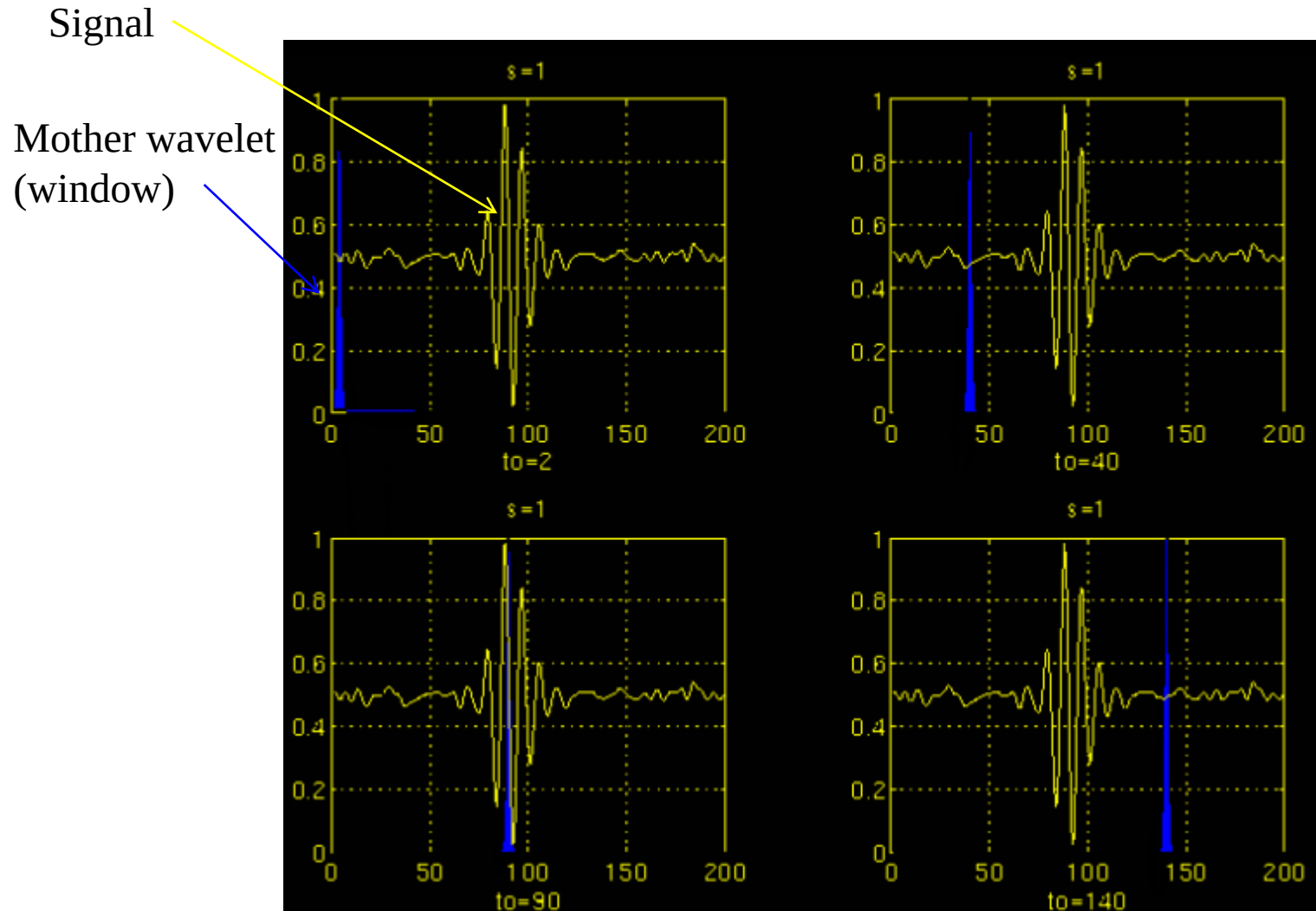
Multi-Resolution Analysis (MRA)

❖ MRA, as implied by its name, analyzes the signal at different frequencies with different resolutions. Every spectral component is not resolved equally as was the case in the STFT.

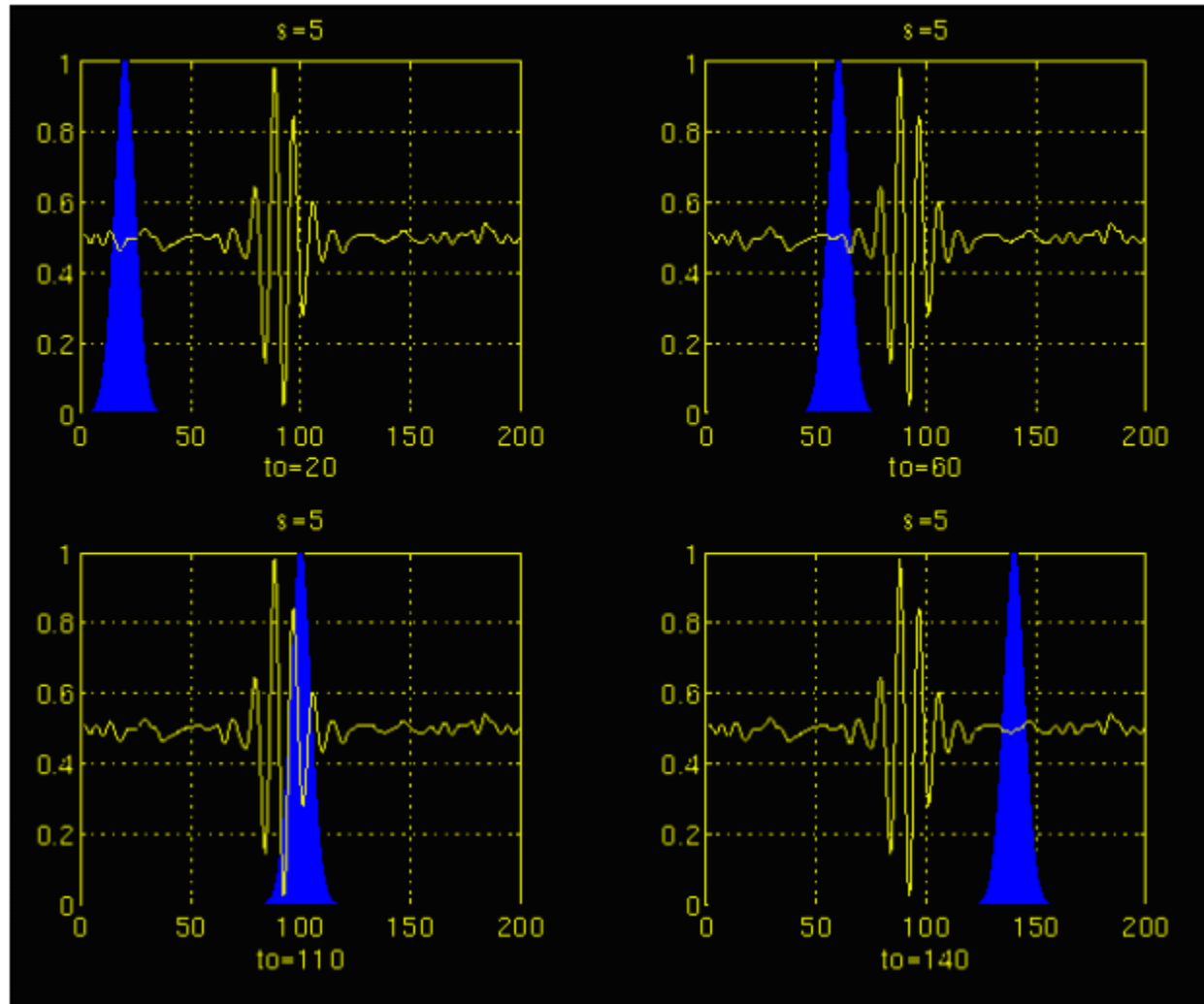
❖ MRA is designed to give good time resolution and poor frequency resolution at high frequencies and good frequency resolution and poor time resolution at low frequencies.

❖ This approach makes sense. Fortunately, the signals that are encountered in Practical applications are often of this type. They have a relatively low frequency component throughout the entire signal and relatively high frequency components for a short duration some where around the middle.

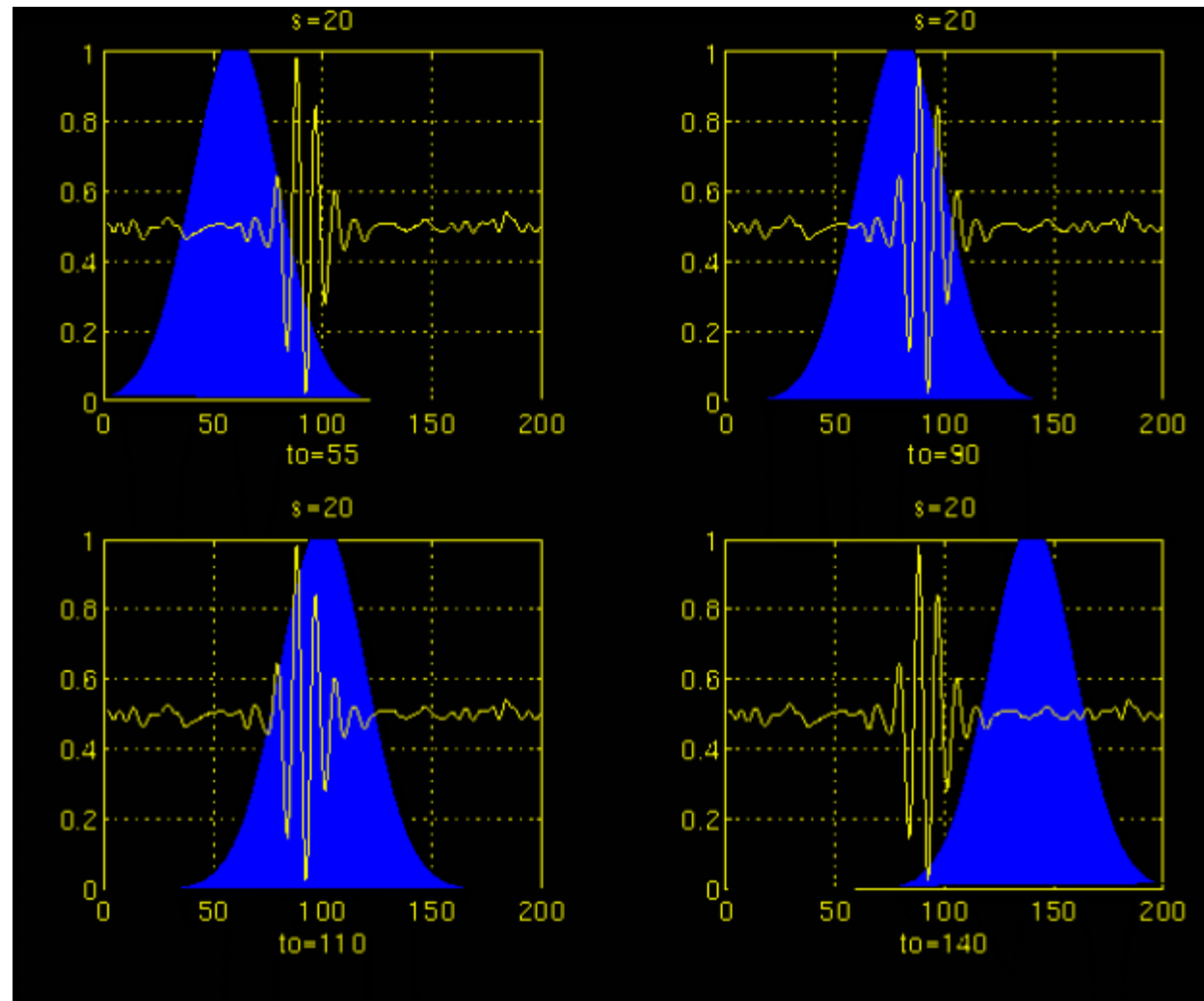
WT algorithm (example)



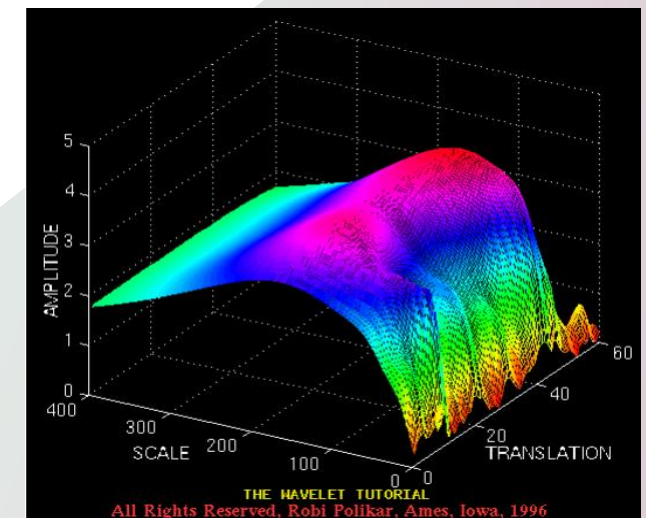
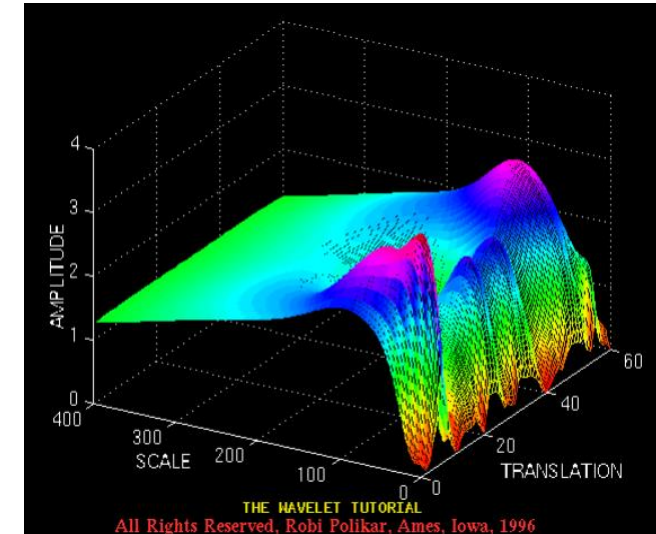
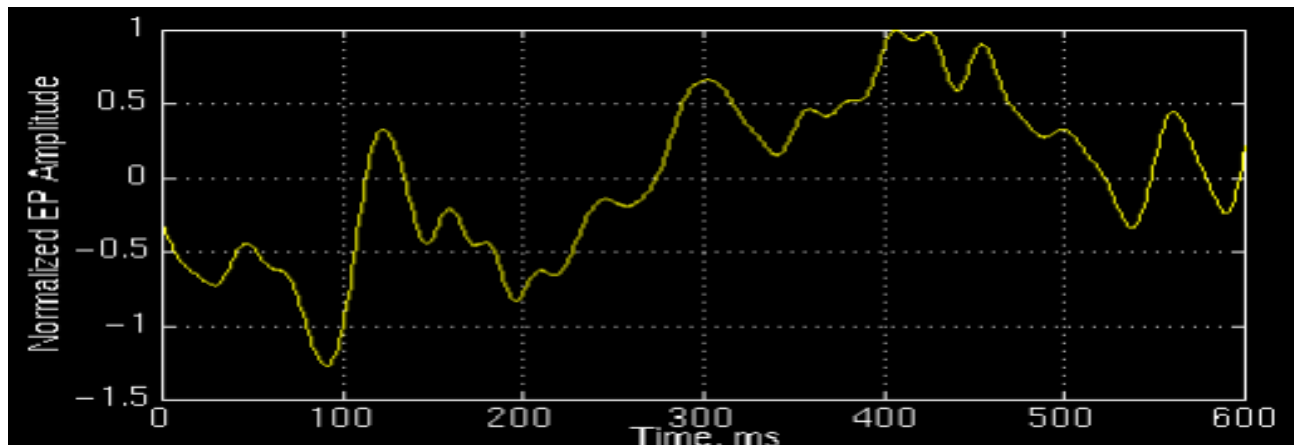
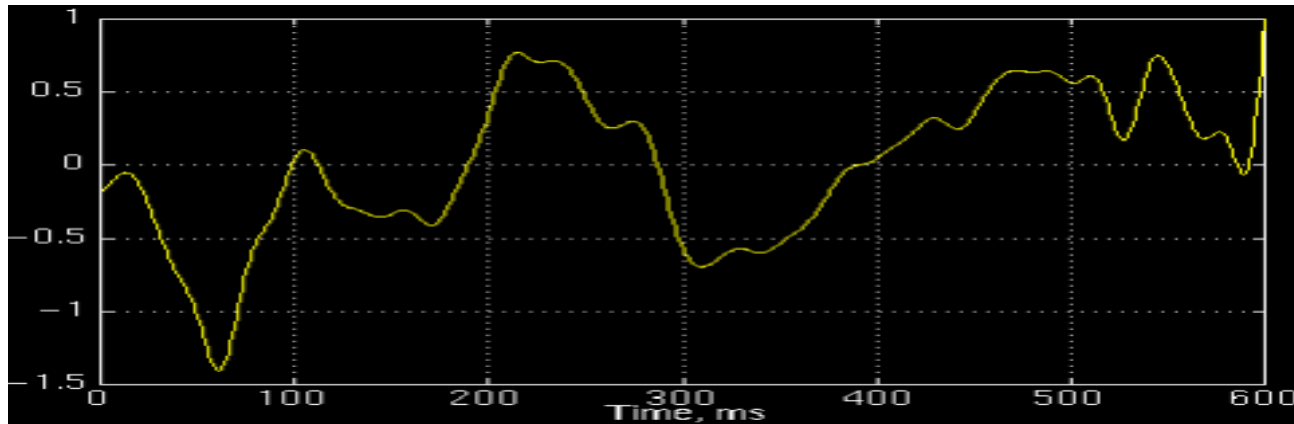
WT algorithm (example)



WT algorithm (example)

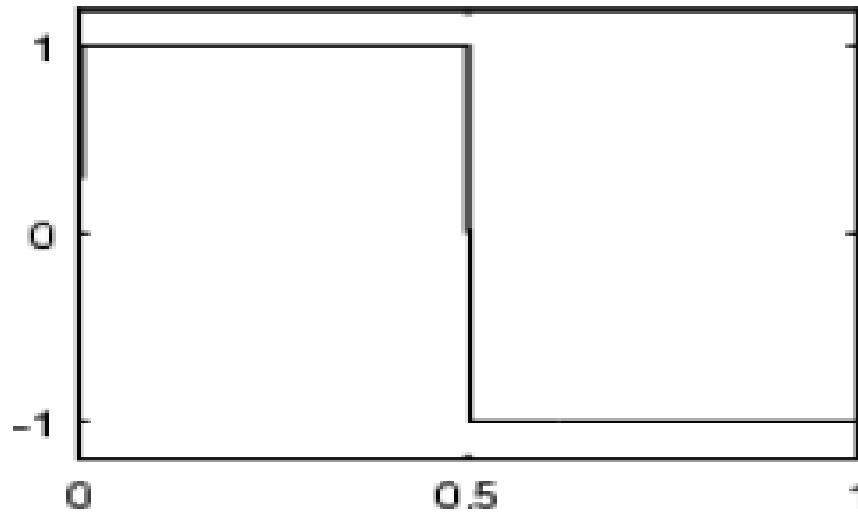


WT algorithm (example)



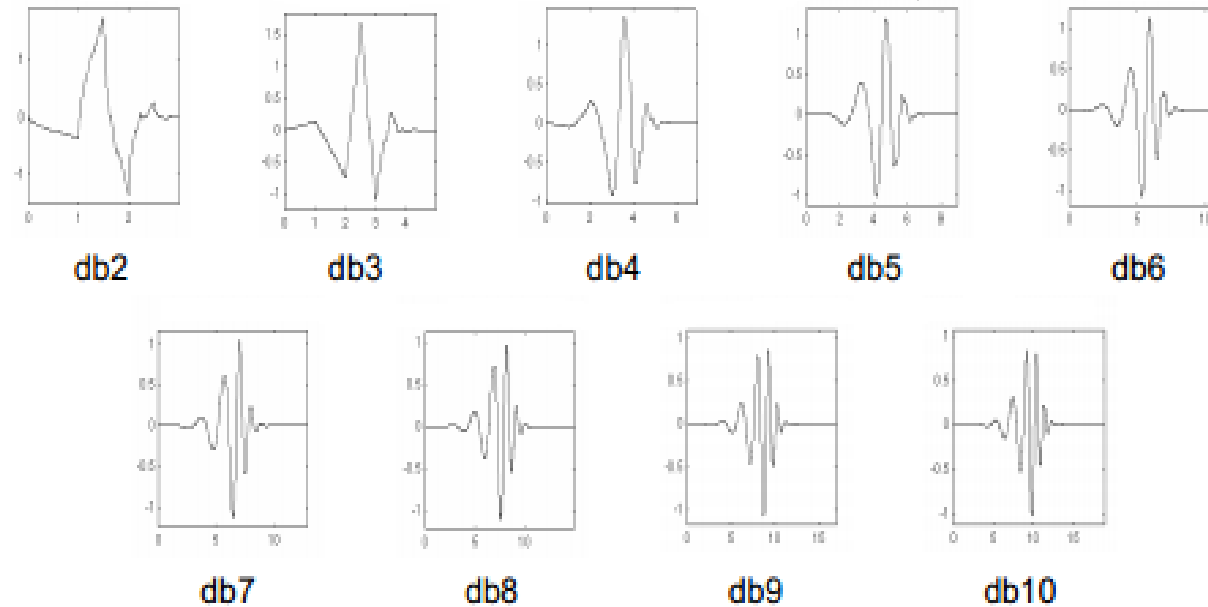
Wavelets Families

The first and simplest wavelet in the world of wavelet transform is Haar wavelet.



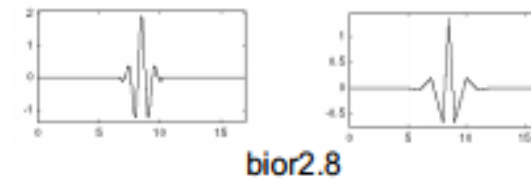
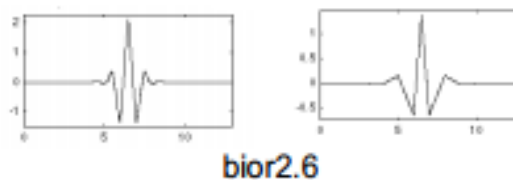
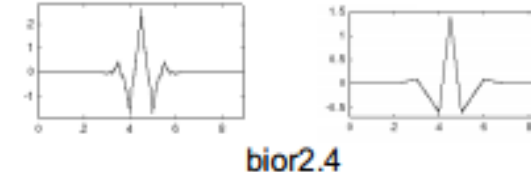
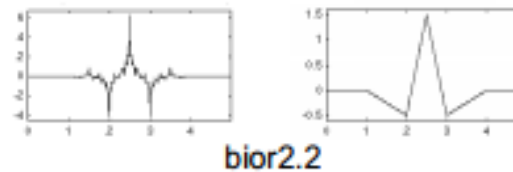
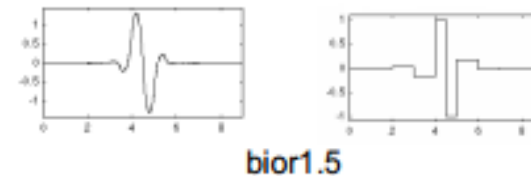
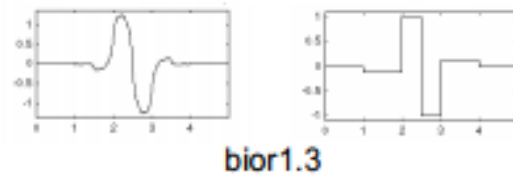
Wavelets Families

**Daubechies wavelets family is the most popular family of wavelets .
Db1 is haar wavelet.**



Wavelets Families

Biorthogonal family of wavelets exhibits the property of linear phase, which is needed for signal and image reconstruction.



Discrete Wavelet Transform (DWT)

- ❖ Although the discretized continuous wavelet transform enables the computation of the continuous wavelet transform by computers, it is not a true discrete transform.
- ❖ the wavelet series is simply a sampled version of the CWT, and the information it provides is highly redundant as far as the reconstruction of the signal is concerned. This redundancy, on the other hand, requires a significant amount of computation time and resources.
- ❖ The discrete wavelet transform (DWT), on the other hand, provides sufficient information both for analysis and synthesis of the original signal, with a significant reduction in the computation time.

Discrete Wavelet Transform (DWT)

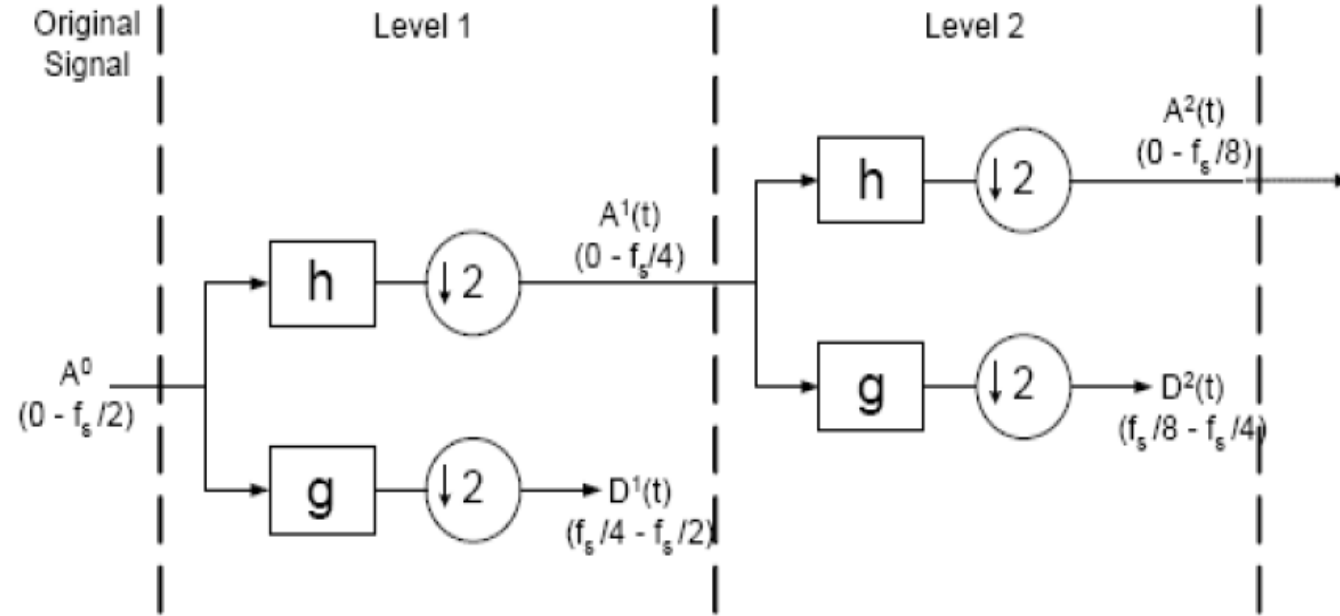
❖ The DWT procedure starts by passing the original signal through a low-pass filter with impulse response of $h[n]$. The low-pass filter, removes all frequencies greater than the cut off frequency of the filter. The signal also passes through a high-pass filter with impulse response of $g[n]$. The high-pass filter removes all frequencies less than the cut off frequency of the filter.

❖ In the wavelet analysis, low frequency components of the signal obtained by low-pass filtering are called approximations and high frequency components obtained by high-pass filtering are called details.

Discrete Wavelet Transform (DWT)

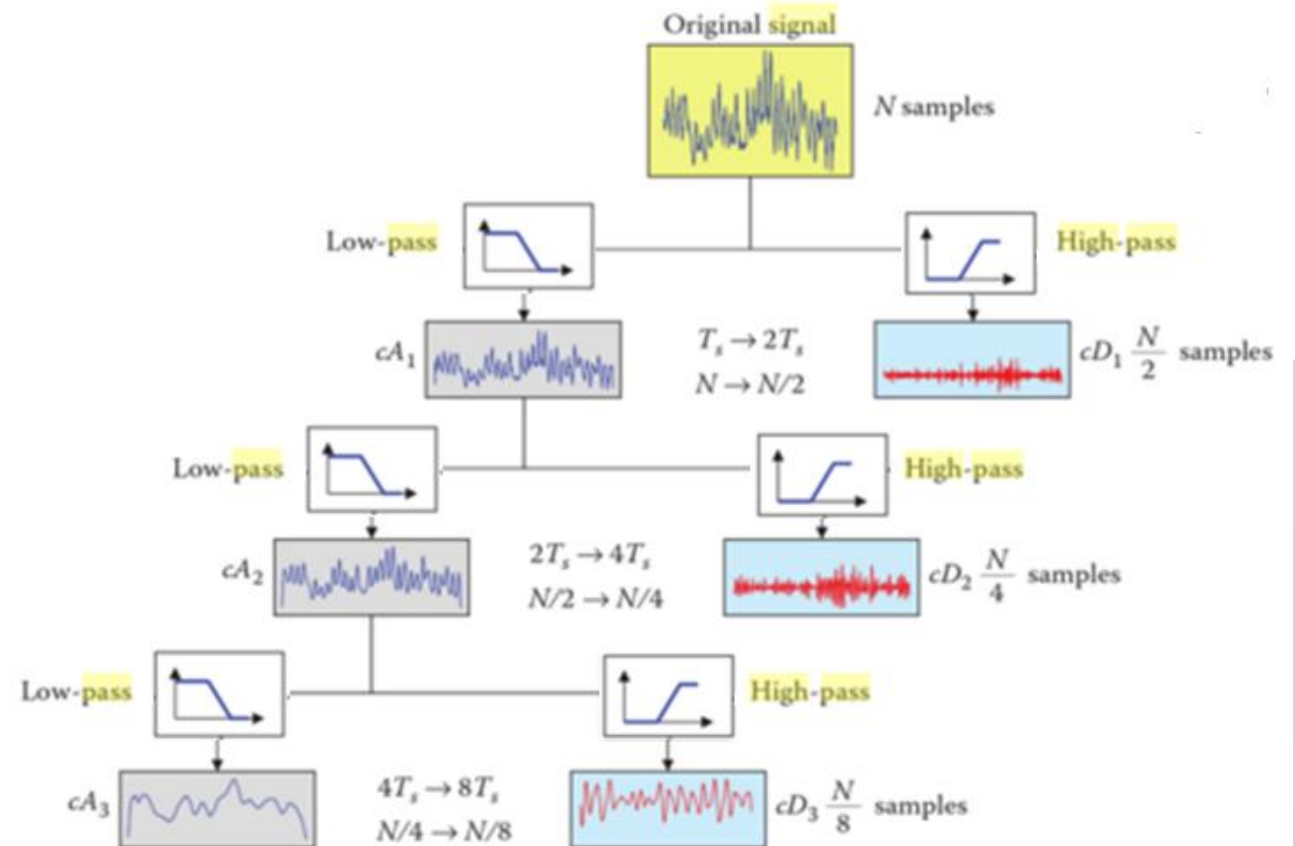
- ❖ Two times filtering the original signal gives twice as much data as the original signal; so, to reduce the number of obtained samples, after filtering the signal down-sampling is a subtle way.
- ❖ The above decomposition procedure can be repeated for further level decomposition. The second level decomposition is obtained by low pass and high pass filtering of the level one approximation of the original signal.
- ❖ Since the MRA analysis process is iterative, in theory it can be continued indefinitely. The decomposition can proceed only until the individual details consist of a single sample or pixel. In practice, you'll select a suitable number of levels based on the nature of the signal.

Discrete Wavelet Transform (DWT)

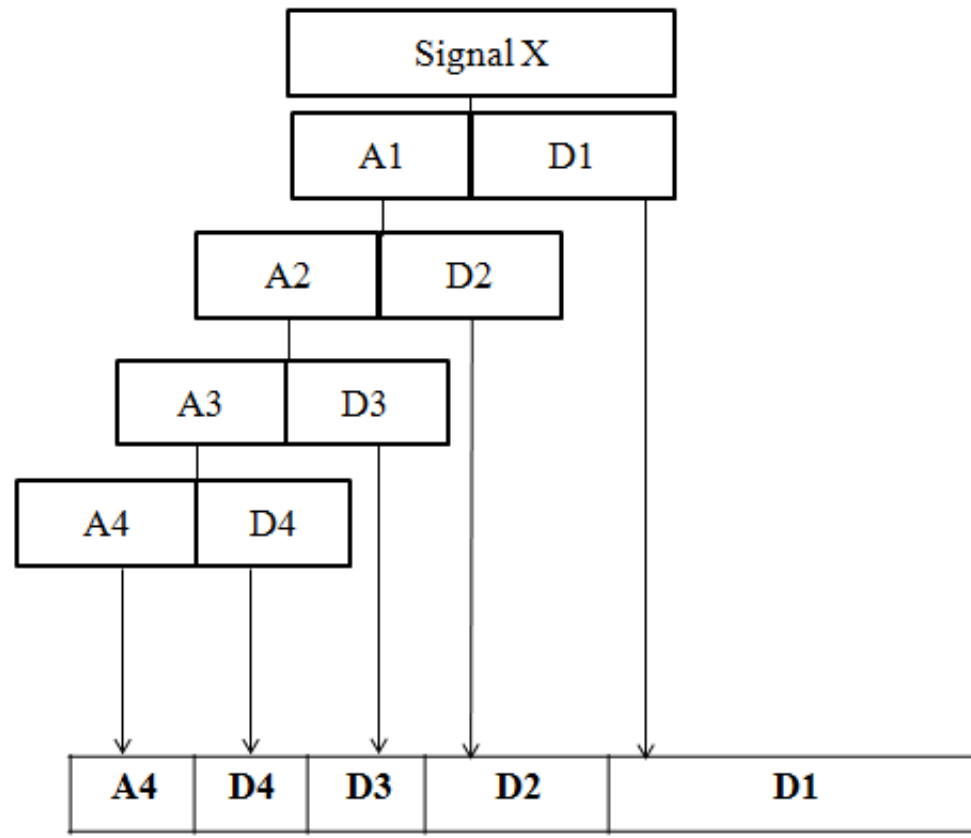


Multi-Resolution Wavelet Decomposition. h = low-pass decomposition filter; g = high-pass decomposition filter; $\downarrow 2$ = down-sampling operation. $A_1(t)$, $A_2(t)$ are the approximated coefficient of the original signal at levels 1, 2 etc. $D_1(t)$, $D_2(t)$ are the detailed coefficient at levels 1, 2 etc.

Discrete Wavelet Transform (DWT)

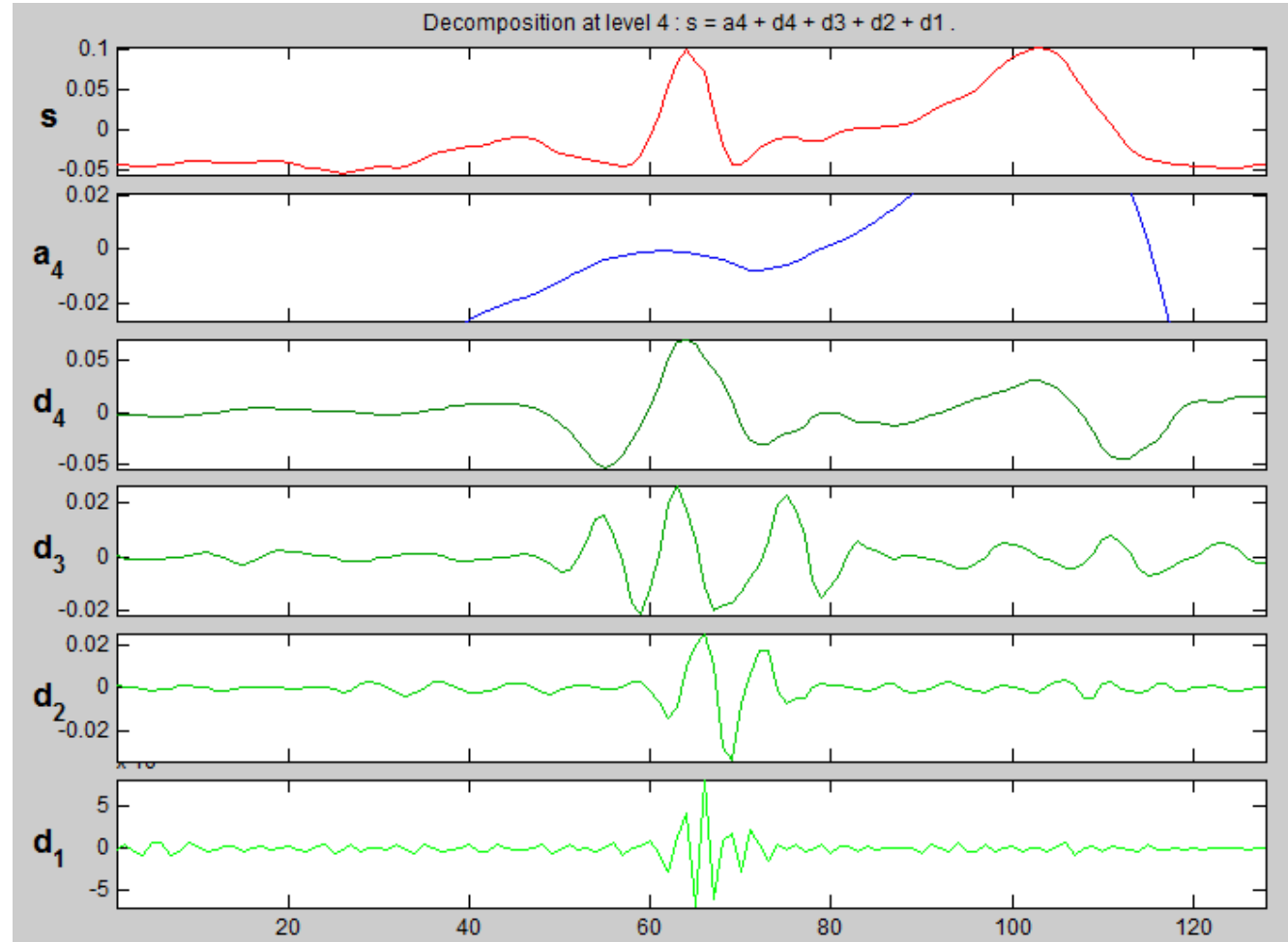


Discrete Wavelet Transform (DWT)

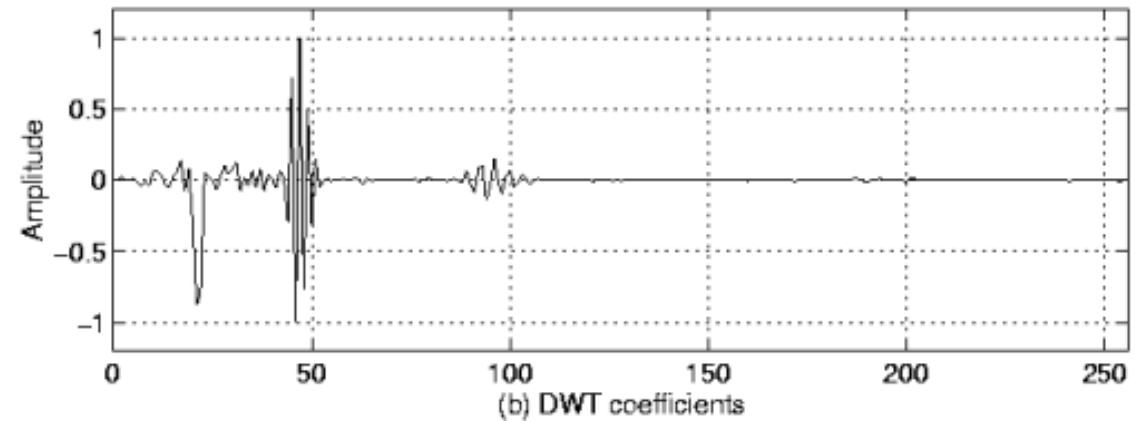
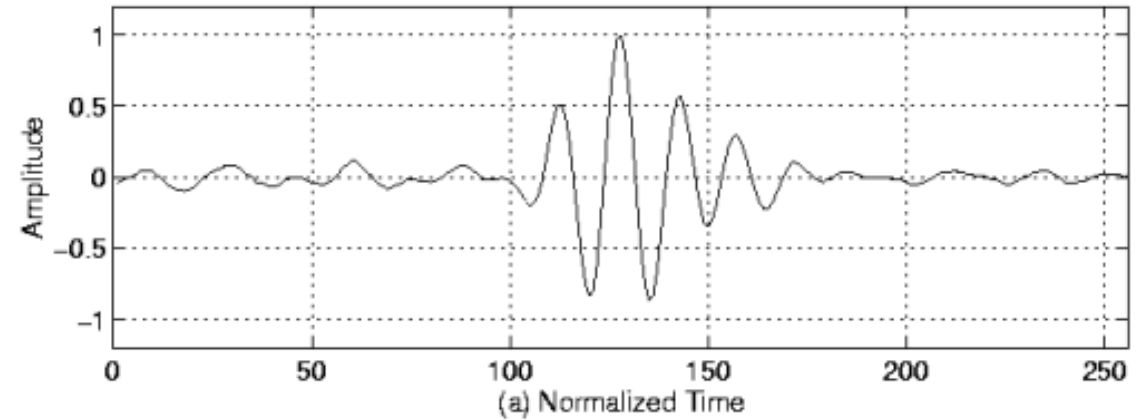


Wavelet coefficient structure

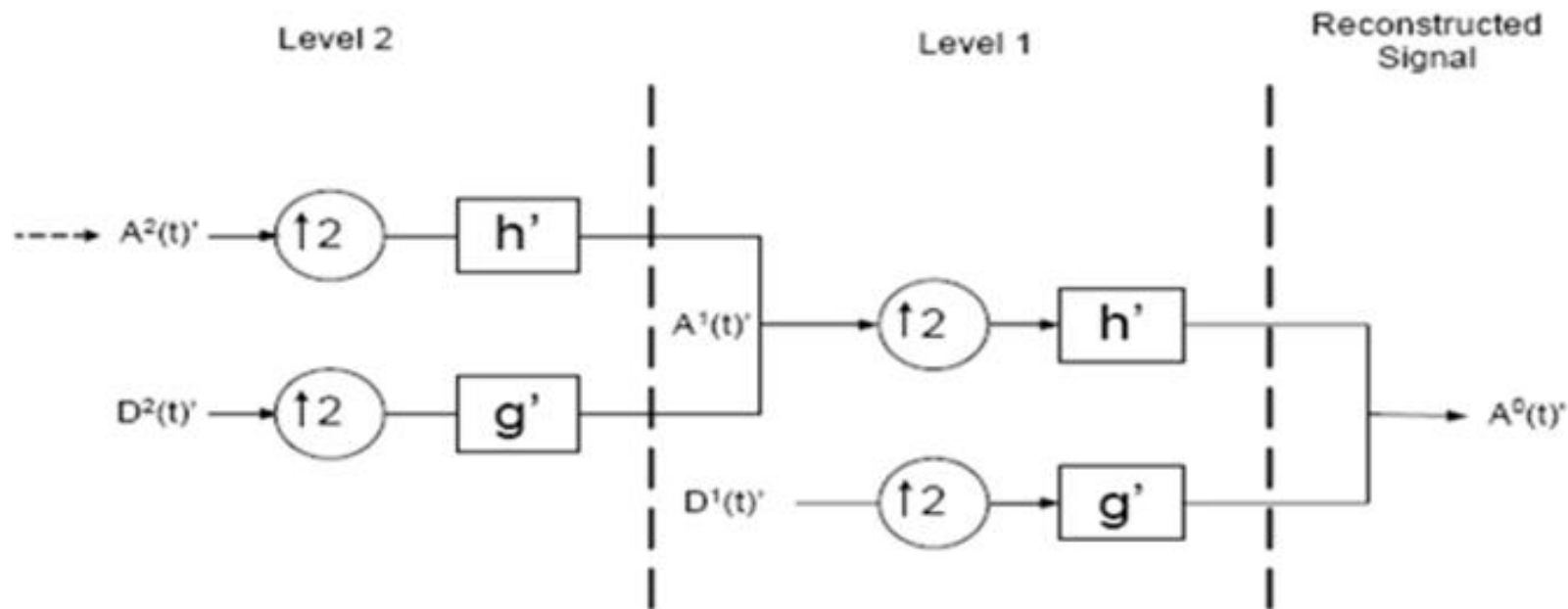
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Multi-Resolution Wavelet Reconstruction. h' = low pass reconstruction filter; g' = high pass Reconstruction filter; $\uparrow 2$ = up-sampling operation. $A_1(t)'$, $A_2(t)'$ are the processed or non-processed approximated coefficient of the original signal at levels 1, 2 etc. $D_1(t)'$, $D_2(t)'$ are the processed or non processed detailed coefficient at levels 1, 2 .

Enhancing Signals in Time and Frequency Domains



Enhancing Signals in Time and Frequency Domains

- ❖ Smoothing or Sharpening
- ❖ Noise reduction or Band selection
- ❖ Squaring
- ❖ DC component removal
- ❖ Resampling
- ❖ Normalization

Smoothing & Sharpening

- **Low Frequencies: Slow small change in values**
signal → smoothing image → blurring

Smoothing preserves low frequencies

- moving average (each value is the average of its surroundings)
- low pass filter

- **High Frequencies: Sudden noticeable change in values**
signal → includes sharp peaks image → includes sharp edges

Sharpening preserves high frequencies

- Derivatives
- high pass filter

Noise Reduction or Band Selection

➤ Filtering

If removing frequencies below specific value is needed, then use high pass filter with cut off frequency equals to this value.

If removing frequencies above specific value is needed, then use low pass filter with cut off frequency equals to this value.

If removing frequencies below & above specific values is needed, then use band pass filter with cut off frequencies equal to these values.

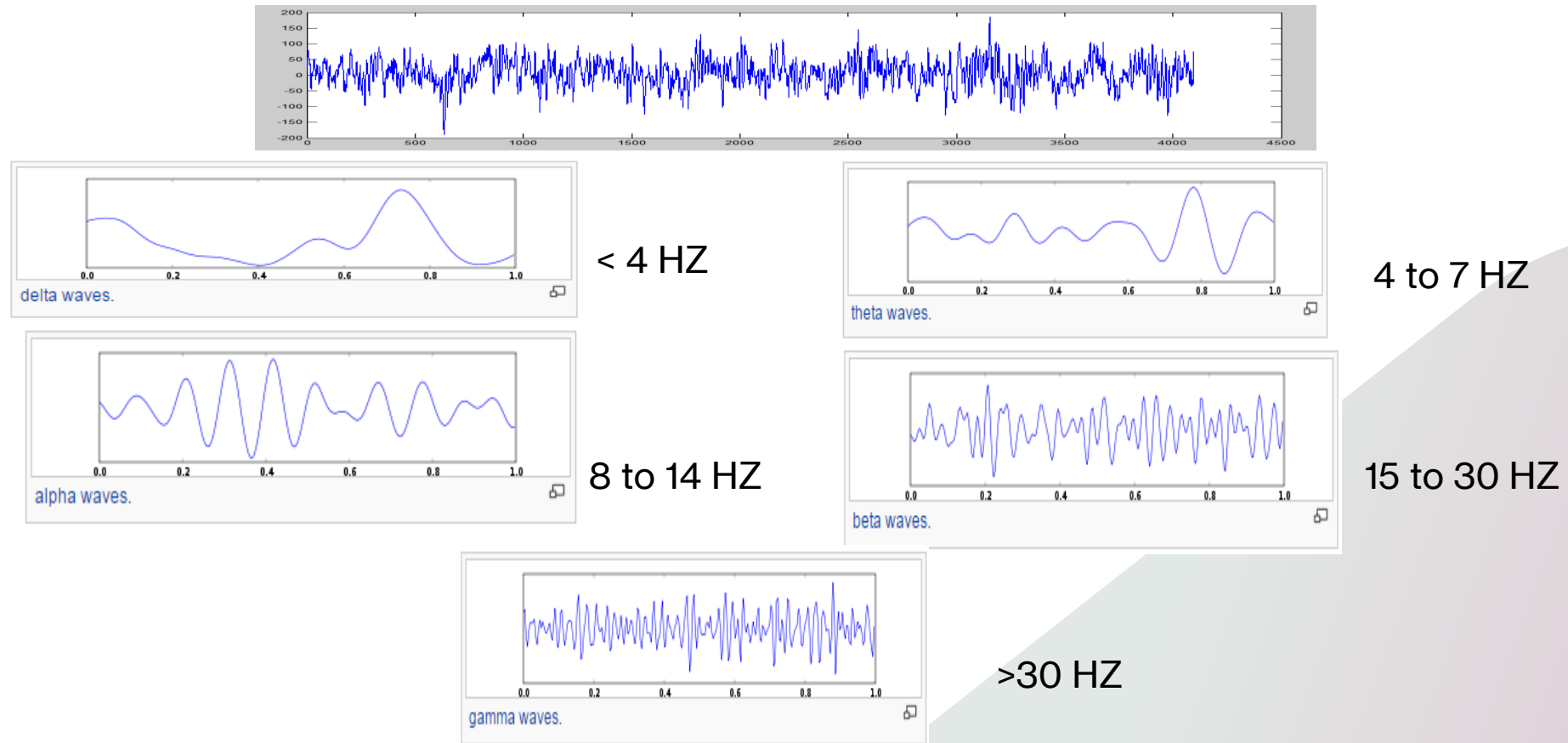
If removing specific band is needed, then use band reject filter with cut off frequencies equals to the band limits.

Noise Reduction or Band Selection

➤ Wavelet Analysis

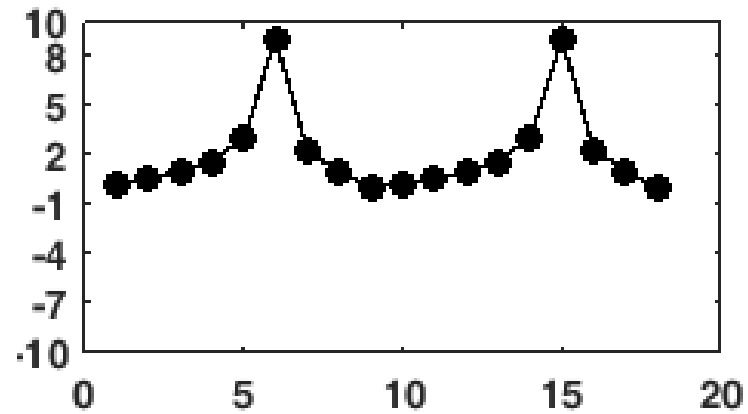
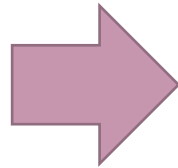
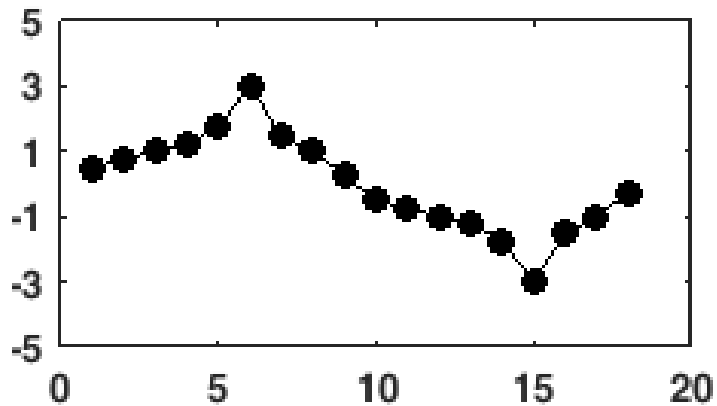
- Digital Wavelet decomposition to suitable number of levels using specific mother wavelet.
- From the resulted wavelet coefficient structure, choose the parts (details or final approximation) that includes the needed band.
- If the signal is needed in time domain, reconstruct the signal (IDWT) using the chosen parts only.

Noise Reduction or Band Selection (example)



Squaring

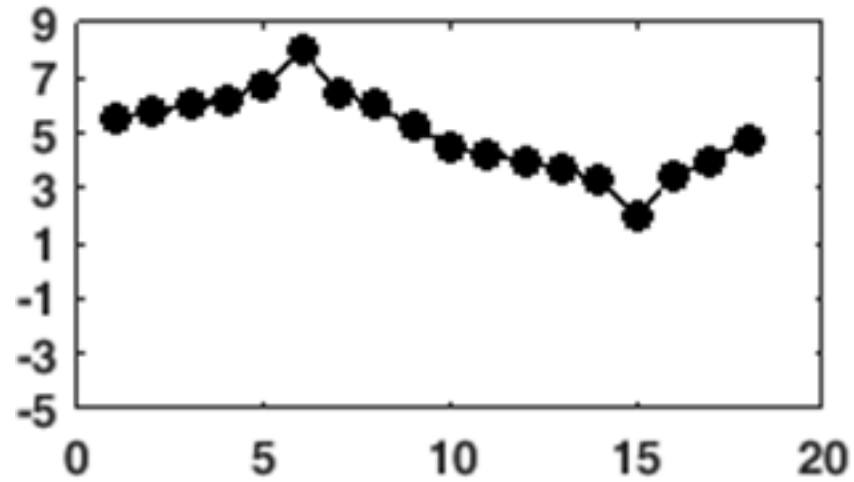
- All values positive
- Increase differences between values (peaks)



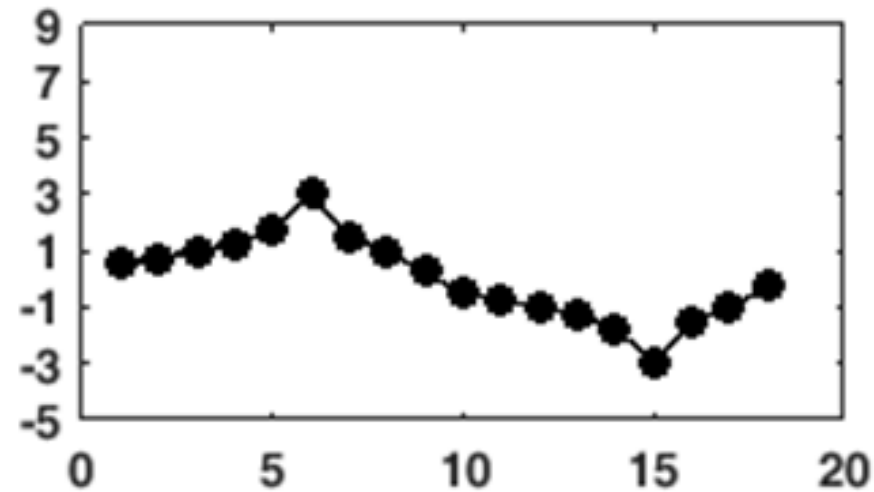
DC Component Removal

- The DC component if it is not equal zero, it makes the signal oscillates on a horizontal line rather than the X axis. Thus, the signal is shifted upwards or downwards.
- We are usually interested in removing the shift and allow the signal to oscillate back on the X axis.
- For time Domain Subtract the mean of the signal from it.
- For Frequency domain remove the first harmonic $X(0)$.

DC Component Removal



DC equals 5

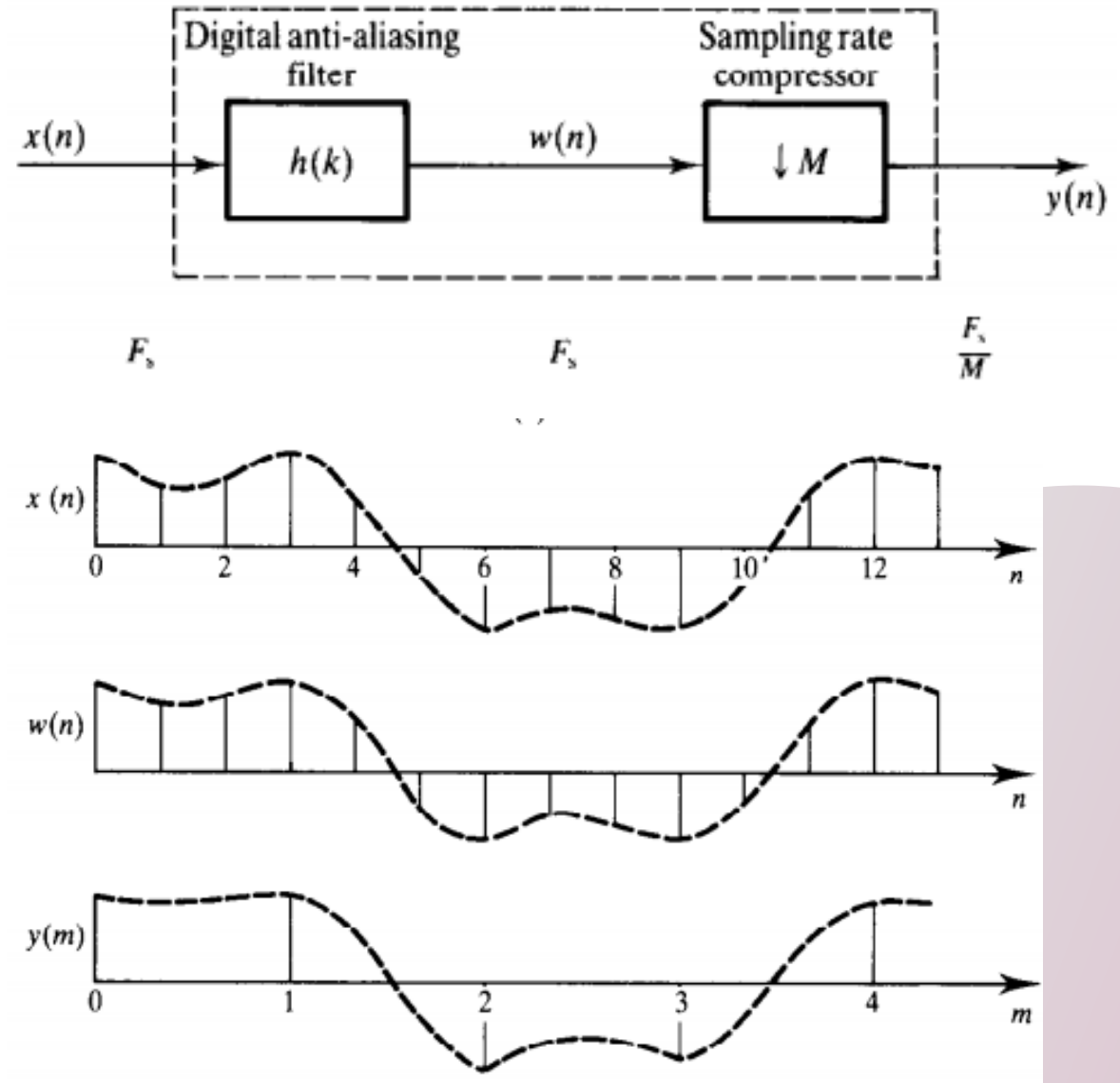


After removing
DC

Resampling (decimation by value)

Take sample and skip M-1

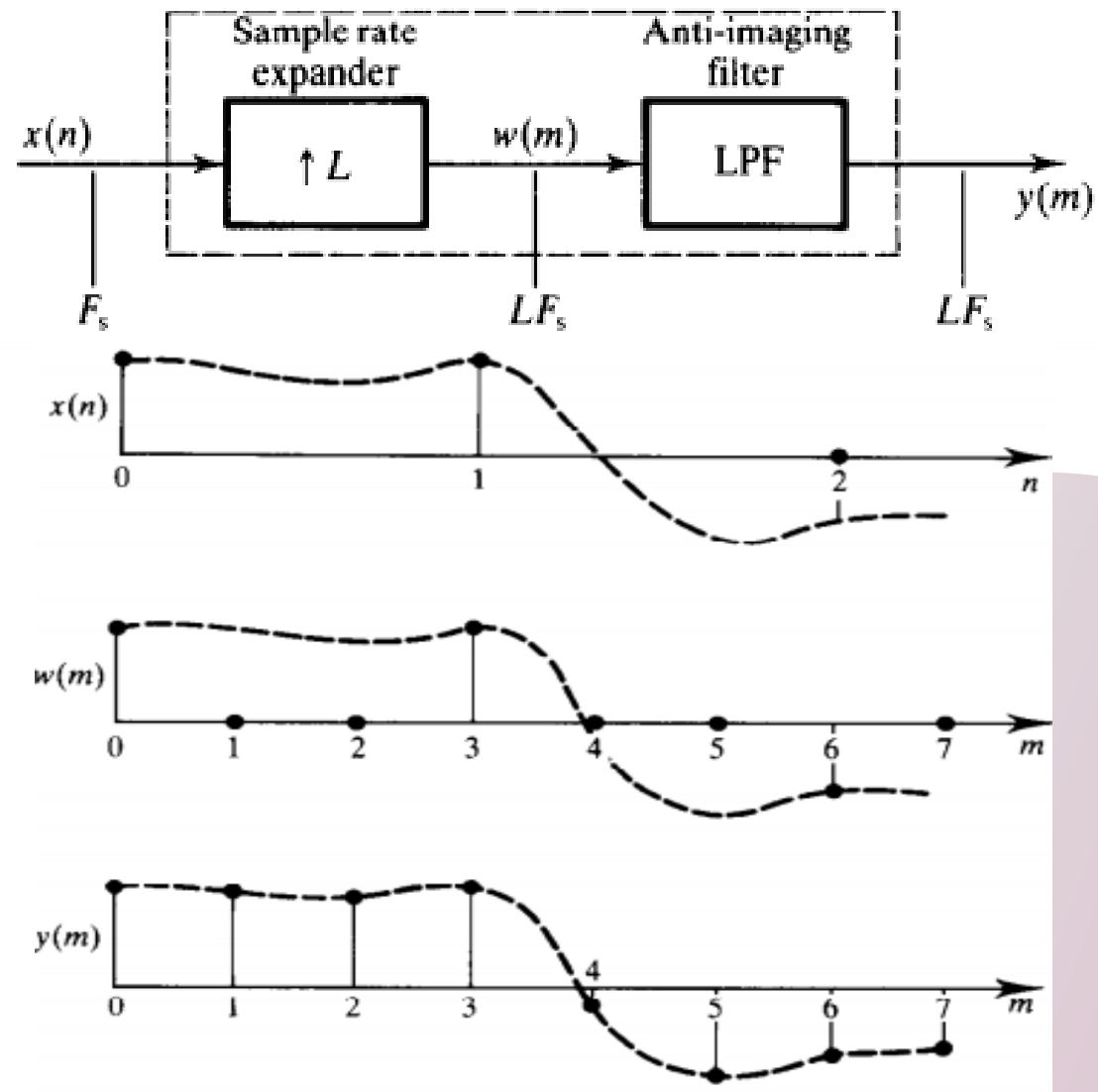
M=3



Resampling (interpolation by value)

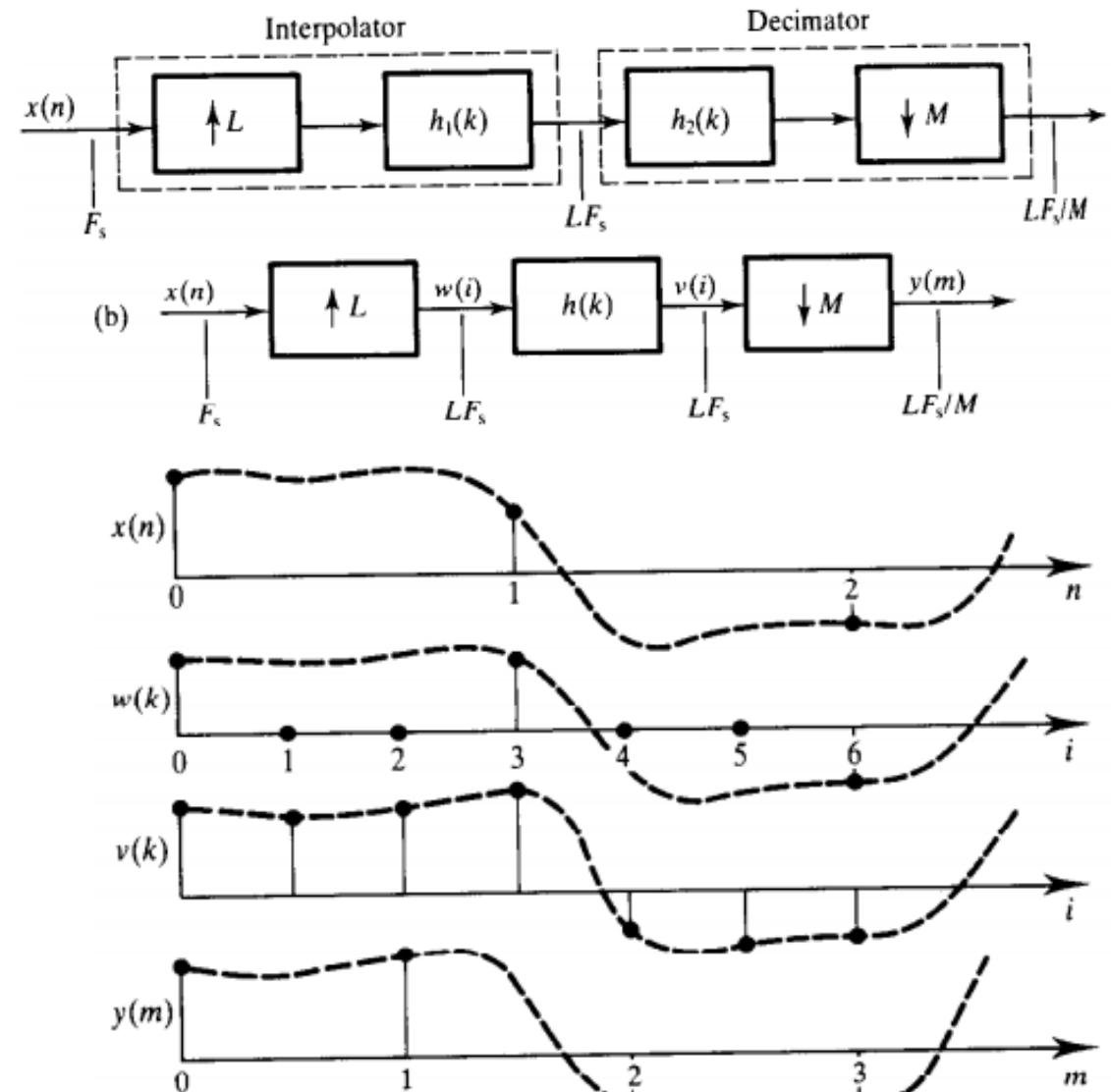
Add $L-1$ zeros between each two samples

$L=3$



Resampling (non-integer values)

Resample by L/M : Add $L-1$ zeros between each two samples, low pass filter to smooth the signal and avoid aliasing and then take sample and skip $M-1$



rational factor ($L = 3, M = 2$).

Normalization

the goal of normalization is to change the values of numeric columns or rows in the dataset to a common scale, without distorting differences in the ranges of values.

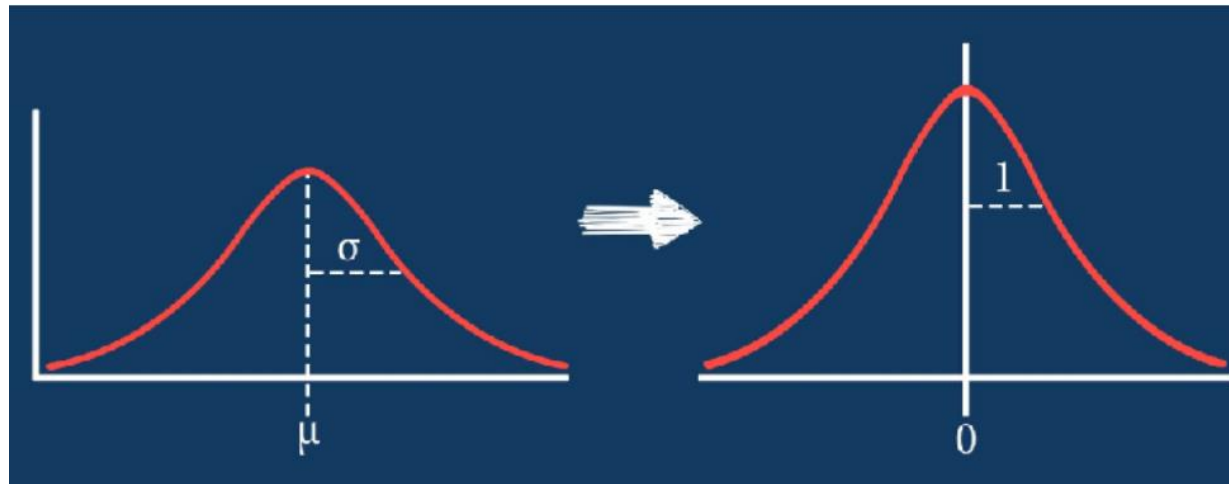
$$X_{\text{new}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad \rightarrow \quad [0, 1]$$

$$X_{\text{new}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad \rightarrow \quad [-1, 1]$$

Normalization

a dataset can be also normalized by subtracting its mean and then divide by its standard deviation

$$X_{new} = \frac{X - \mu}{\sigma}$$



Thank you